

## Descending Pathways to the Spinal Cord: A Comparative Study of 22 Mammals

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### ABSTRACT

In order to estimate the qualitative commonalities and range of variation among major descending spinal pathways relevant to mankind's ancestral lineage, the supraspinal cell groups originating fibers descending directly to the spinal cord were examined in 22 mammalian species. In a standardized retrograde tract-tracing procedure, flakes of raw HRP were applied directly to the freshly cut fibers of the spinal cord after it had been hemisected at the C1-C2 junction. After a 72-hour survival period, brain and spinal cord tissues were processed by conventional HRP-processing techniques.

This procedure was performed on 94 individual animals. Of this total, 41 individual cases were eliminated by a rigorous culling procedure. The results are based on 53 individuals representing 15 species selected for their successive kinship with mankind and seven species in two other lineages selected for the convergence of their visual or sensorimotor systems with anthropoids. The 22 species represent 19 genera, 14 families, eight orders, and two subclasses of Mammalia.

The results show that at least 27 supraspinal cell groups, each containing intensely labeled cells, can be readily identified in each of the species. Despite vast quantitative differences in cell number and cell size, this qualitative uniformity among the relatively large number of diverse taxa suggests that the same pathways were probably present in the extinct ancestors throughout mankind's mammalian lineage and are probably still present in extant viviparous mammals as well. If so, these pathways are as old in phylogenetic history as the last common ancestor of marsupial and placental mammals—dating from the late Jurassic to early Cretaceous, perhaps 145–120 million years ago.

Further comparison of the results with similar experimental findings in members of other vertebrate classes supports the notion that several of these same pathways can be traced to even more remote ancestry, with some possibly as old as the entire vertebrate subphylum—dating from the early Devonian or before, perhaps 430 million years ago. Within mankind's ancestral lineage, from the appearance of vertebrates to the appearance of mammals, there seems to have been an irregular stepwise augmentation of the set of descending pathways until the full mammalian complement was finally attained with the appearance of the corticospinal tract.

**Key words:** motor systems, spinal cord afferents, efferent tracts

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Perhaps because of the availability of reliable orthograde tract-tracing techniques for over a century (Marchi, 1886; Nauta and Gyax, '54; Cowan et al., '72), most detailed accounts of brain evolution have consisted largely of the history of the sensory systems and closely allied structures (e.g., Petras and Noback, '69; Pearson and Pearson, '76; Masterton et al., '76; Sarnat and Netsky, '74). With the

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discovery and development of equally reliable retrograde tract-tracing techniques, however (Kristensson and Olsson, '71; LaVail and LaVail, '72; Mesulam, '78), this sensory-centered view of comparative and evolutionary neurology has begun to change rapidly. The sites of origin, trajectories, and terminations of motor and premotor pathways are now beginning to be described in a widening variety of vertebrates.

Experimental studies illustrating the supraspinal origins of pathways descending to the spinal cord are already available for many individual species. Although each of these studies is useful in its own right, large procedural differences among the studies have tended to preclude their detailed comparison, often obscuring the similarities and differences among the animals. Consequently, the array of useful comparative data remains far too small to provide more than conjecture concerning even the presence or absence of particular tracts.

The present report is the first of a series focused on descending pathways in mammals. The goal of the experiments was to provide an array of histological materials from a relatively large number of species, all prepared with virtually identical surgical and histological techniques, so that both qualitative and quantitative comparisons across species could be more safely made. With most of the experimental animals selected primarily on the basis of their relative recency of common ancestry with mankind, it was also hoped that these materials might be useful for augmenting present views of brain evolution.

We began the series with strictly qualitative studies in order to establish first the ubiquity or rarity of particular descending tracts already described by others as present in one or another mammal. Later, we progressed to quantitative studies of several of the longest common tracts (tectospinal, hypothalamospinal, corticospinal) together with statistical analyses exploring their relationship both to the ancestry and the current ecology of their hosts.

In the present report, we have used conventional histoarchitectonic, cytoarchitectonic, and hodological criteria in an attempt to determine the qualitative commonalities and possible differences among 22 species.

## MATERIALS AND METHODS

In order to provide assurance that the histological materials would maximize retrograde labeling, we began with a series of quantitative experiments on laboratory rats. In these preliminary experiments, we compared the efficacy of several retrograde-tracing methods, systematically varying tracers, survival times, perfusates, chromagens, and other histological parameters. For each variation of method, we estimated the total number of labeled cortical and tectal cells by counting the labeled cells in a systematic sample of sections and then correcting the counts with appropriate stereological methods (Weibel, '79). These corrected counts were then compared among the several procedures and their respective modifications and also with counts from similar experiments published by others. The retrograde-labeling methods yielding lower counts were gradually eliminated one by one and those yielding the higher counts were retained. This procedure continued until only one method remained—the one yielding the highest count in every case.

The technique eventually adopted as the standardized procedure for the experiments was the horseradish peroxi-

dase bulk-filling procedure of Hicks and D'Amato ('77) with tetramethyl benzidine (TMB) processing (Mesulam, '78). In our hands at least, no other procedure ever resulted in higher numbers of retrogradely labeled cortical or tectal neurons. This procedure had the additional feature of being usable without variation for each of the several species.

In our standardized version of this procedure, flakes of raw HRP were applied directly to the freshly cut fibers of the spinal cord after it had been hemisected at the C1-C2 junction. After a 3-day survival period, the resulting materials were processed with tetramethyl benzidine.

## Surgical procedure

The surgical and histological procedures were virtually identical for each animal. First, the animal was anesthetized with 3.3 cc/kg of a mixture of chloral hydrate and pentobarbital (Chloropent, Fort Dodge). Through a dorsal approach, the cervical neck muscles were protracted from the midline to expose the atlas and axis. After cutting and then retracting the dura between atlas and axis, the dorsal roots of the second cervical nerve were located. Avoiding the vertebral arteries, parts of the atlas were usually removed, exposing a small area of cord rostral to the dorsal roots of C2. The cord was then hemisected with small scissors on the right side just rostral to the C2 dorsal roots (Fig. 1).

The choice of the C1-C2 level for the filling of axons descending to the spinal cord was a compromise between two conflicting ideals. On one hand, the primary goal was to label the entire complement of fibers descending to the spinal cord; therefore, the ideal level for the hemisection was the medulla-spinal cord junction. On the other hand, the unavoidable spread of extraaxonal HRP could not be allowed to extend to terminal fields in lower bulbar structures (e.g., the dorsal column nuclei) without contaminating the results. On the basis of the preliminary experiments, the C1-C2 level proved to be the highest-level site of HRP application that could be unequivocally identified in the diverse species and, at the same time, markedly reduced (but did not eliminate) the chance of the marker spreading into the medulla. Despite this precaution, however, the subsequent histological procedures showed that in a few individual cases the marker did indeed spread far enough upward to intrude on the dorsal column nuclei. These cases were eliminated from the study entirely and new ones were, when available, prepared to take their place.

Immediately after the hemisection, dry flakes of HRP (Boehringer-Mannheim, grade 1) were applied directly to the cut and reapplied periodically for the next 20–30 minutes. Cottonoid strips placed around the site of application absorbed excess HRP and any cerebrospinal fluid or blood leaking from the wound. After 30 minutes, any excess HRP was carefully absorbed with cottonoid, the cottonoid strips were removed, Gelfoam was placed over the cut, the muscles were sutured at the midline, and the wound was closed. The animal was then treated systemically with ampicillin (6.6 mg/kg) and topically with lidocaine. Although in a few cases doxapram hydrochloride (1 mg/kg) was administered to stimulate smooth respiration, the vast majority of the animals were awake within 2–3 hours and active within 8–10 hours.

At the end of the 72-hour survival period, the animal was injected with a lethal dose of Chloropent and perfused, first with 0.9% saline/0.2% lidocaine hydrochloride/1% heparin at room temperature, then with 1.25% glutaraldehyde/1.0%

paraformaldehyde at 10°C for 30 minutes, and finally with 10% sucrose at 10°C. Each of these perfusates was phosphate buffered at pH 7.4.

The brain and spinal cord were then removed and the hemisection was examined for its adequacy. Following photographic documentation of the level and the extent of the hemisection, the brain and spinal cord were separated approximately 1 mm caudal to the hemisection; then they were immersed in 30% sucrose (pH 7.4) and stored at 4–10°C. After the tissue had sunk (1–2 days), it was weighed and embedded in gelatin. The gelatin blocks were fixed in 2% glutaraldehyde overnight at 4–10°C. The next day the brain was cut in 50- $\mu$ m frontal or sagittal sections. The sections were collected in cryoprotectant (de Olmos et al., '78), and stored at –10°C until processing.

### Histological procedure

The HRP-processing procedures were carried out within 7 days of death for most of the tissues and for all of the tissues used for quantitative measurements. The tissue was divided into five sets of sections, each set representing one-fifth of the total.

Set 1 was processed according to the TMB procedure of Mesulam ('78) at room temperature for 20 minutes. Set 2 was processed according to the same procedure but at 4–10°C for 25 minutes. Set 3 was processed according to the same procedures at 4–10°C for 25 minutes and then counterstained with cresyl violet. Set 4 was processed according to the same procedure at 4–10°C for 25 minutes and then counterstained with neutral red. Set 5 was reserved for further experimental variations in the histological procedure in attempts to increase the total number of visibly labeled neurons without significantly increasing background artifact. Some of these experimental variations included the TMB procedure of de Olmos et al. ('78), the paraphenylenediamidine-pyrocatechol procedure (Hanker et al., '77), a cobalt-intensified diaminobenzidine procedure (Adams, '81), and various modifications of the TMB procedure suggested by others. Once more, none of these modifications proved to yield higher counts of labeled corticospinals or tectospinals than the two routine TMB procedures (set 1 or set 2 above).

### Identification of labeled cell groups

The identification of the labeled cell groups was based on light microscopic examination mostly of the HRP-processed and counterstained tissue. However, because the labeling method usually resulted in Golgi-like labeling of somata and in many cases, of dendrites and individual axons as well, the uncounterstained materials allowed an unusual number of extra criteria to be used for purposes of nuclear identification.

The criteria for identifying the labeled cell groups included their absolute location; their location relative to neighboring cell groups and fiber tracts; and the absolute and relative size, packing density and specific type of labeled somata; in many cases, the criteria also included the initial axon trajectories and the size, form, and orientation of their dendritic fields; symmetries and asymmetries of labeling; and staining intensity, among others. Cytoarchitectonic brain atlases of several of the species also served as useful guides to identification (e.g., Taber, '61; Emmers and Akert, '63; Berman, '68; Oswaldo-Cruz and Rocha-Miranda, '68; Travis, '82), as did prior reports of descending

pathways in some of the same species (e.g., Kuypers and Maiskey, '75; Burton and Loewy, '77; Basbaum and Fields, '79; Martin et al., '81; Mizuno et al., '85).

### Species represented

In all, the standardized surgical and histological procedures were performed on 94 individual animals. Of this total, the results from 53 animals representing 22 species in 19 genera are reported here. These 19 genera, in turn, represent 14 families, eight orders, and two subclasses of Mammalia (Table 1).

The other 35 representatives of the same set of species and six representatives of two additional species (least shrew and rock hyrax) were eliminated from the study because the spinal cord hemisection was incomplete or misplaced, because the animal did not survive for the full 72-hour survival period, because the marker may have spread rostrally into the dorsal column nuclei, or because the HRP reaction product was judged to be inadequate or partly artifactual. Therefore, the 53 individual cases presented here consist only of those that could not be eliminated by this screening procedure.

Several considerations guided our choice of these particular animals. To begin with, 15 of the species were selected on strict paleontological, taxonomic, and comparative anatomical grounds for their sequential kinship, or relative recency of common ancestry, with mankind. This selection allowed the representation of at least four and possibly seven grades of successive kinship with humans: marsupial, insectivore, prosimian primate, and Old World monkey. To these, armadillo was added on the chance that edentate propinquity with mankind may prove to be intermediate to marsupials and insectivores; tree shrew was added on the chance that Scandentia propinquity with mankind may prove to be intermediate to insectivores and primates; and common marmoset and squirrel monkey were added on the chance that New World monkey propinquity with mankind may prove intermediate to prosimian and Old World monkeys (e.g., McKenna, '75; Lillegraven et al., '79; Eisenberg, '81).

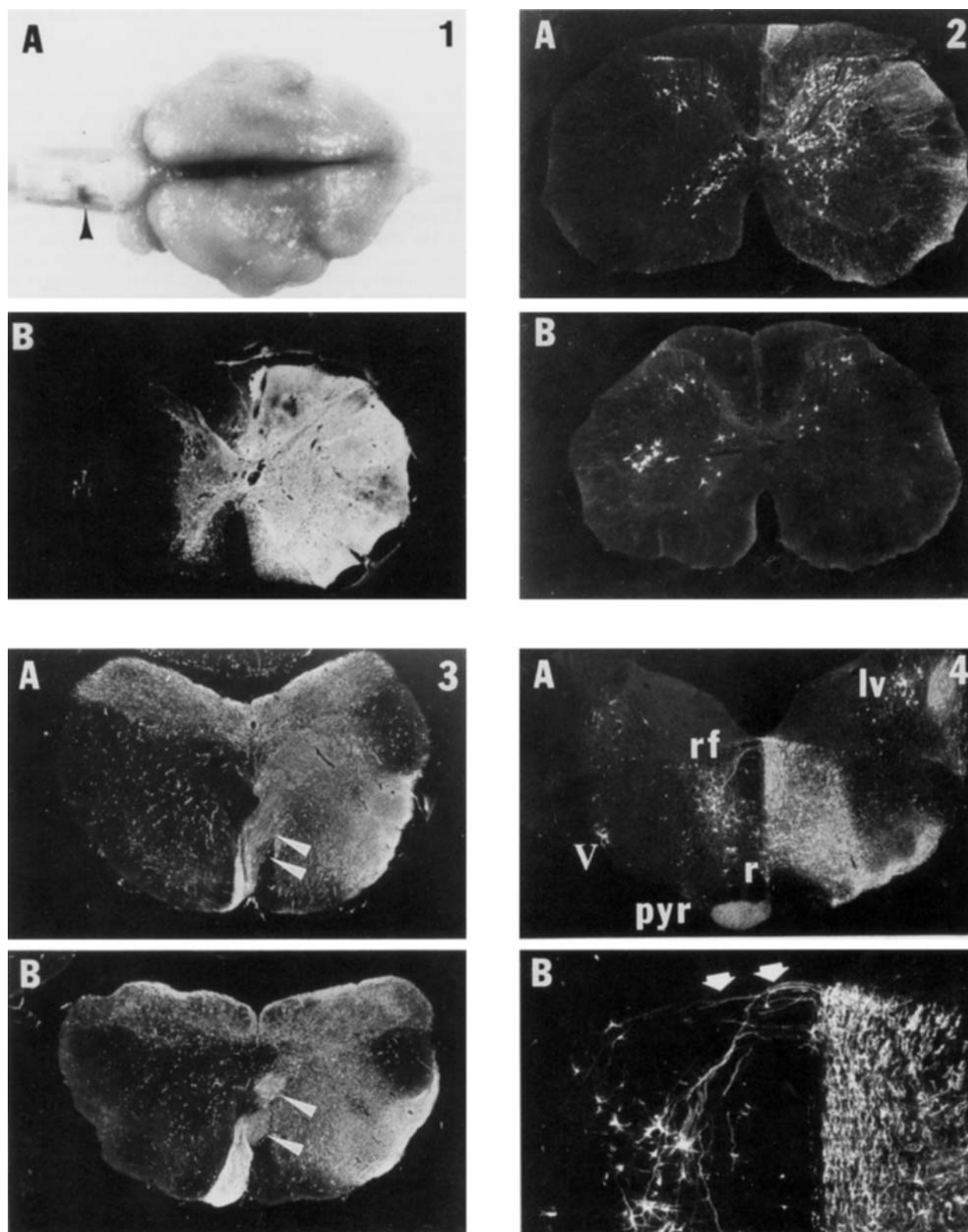
Of the remaining seven species, five were selected from the order of Rodentia and Lagomorpha (rabbit, rat, vole, ground squirrel, tree squirrel) on the basis of their successive kinship with tree squirrels—a lineage superficially paralleling the origin of a prosimianlike visual system.

Finally, two species of Carnivora were selected (cat, raccoon) on the basis of their successive kinship with raccoons—a lineage superficially paralleling the origin of primateline manual dexterity.

## RESULTS

In each of the 53 individual sets of histological materials which survived the screening procedures, large numbers of neurons were vividly labeled throughout the neuraxis from coccygeal spinal cord to cerebral neocortex. The results from a typical case are shown in Figures 1–8 while Figure 9 illustrates the close similarity in cell labeling of two diverse nuclei: the red nucleus (among six different species) and nucleus ambiguus (among four species).

By using the identifying criteria listed above, at least 27 cell groups or complexes, 17 major and ten minor, each containing intensely labeled cells, could be identified in the brain of each individual case. This complement of labeled nuclei could be considerably expanded (to more than 50



Figures 1-4

TABLE 1. Species and Genera Represented

| Common name                | Scientific name                  | No. subjects |
|----------------------------|----------------------------------|--------------|
| North American opossum     | <i>Didelphis virginiana</i>      | 2            |
| Gray, short-tailed opossum | <i>Monodelphis domestica</i>     | 3            |
| Nine-banded armadillo      | <i>Dasypus novemcinctus</i>      | 1            |
| European hedgehog          | <i>Erinaceus europaeus</i>       | 2            |
| West African hedgehog      | <i>E. albiventris</i>            | 2            |
| Desert hedgehog            | <i>Hemiechinus auritus</i>       | 2            |
| Eastern American mole      | <i>Scalopus aquaticus</i>        | 2            |
| Tree shrew                 | <i>Tupaia glis</i>               | 3            |
| Slow loris                 | <i>Nycticebus coucang</i>        | 1            |
| Lesser bushbaby            | <i>Galago senegalensis</i>       | 3            |
| Squirrel monkey            | <i>Saimiri sciureus</i>          | 3            |
| Common marmoset            | <i>Callithrix jacchus</i>        | 4            |
| Green monkey               | <i>Cercopithecus aethiops</i>    | 2            |
| Crab-eating macaque        | <i>Macaca fascicularis</i>       | 2            |
| Rhesus macaque             | <i>M. mulatta</i>                | 1            |
| Domestic cat               | <i>Felis catus</i>               | 2            |
| Raccoon                    | <i>Procyon lotor</i>             | 4            |
| Eastern gray squirrel      | <i>Sciurus carolinensis</i>      | 2            |
| 13-lined ground squirrel   | <i>Citellus tridecemlineatus</i> | 2            |
| Pine vole                  | <i>Pitymys pinetorum</i>         | 1            |
| Rat                        | <i>Rattus</i>                    | 8            |
| Rabbit                     | <i>Oryctolagus cuniculus</i>     | 1            |

separate cell groups) if each of the groups were to be refined into its components (for example, what we call the "bulbar reticular formation" includes paramedian reticular nucleus + gigantocellular tegmental field + magnocellular tegmental field + lateral reticular nucleus while what we call the "raphe complex" includes nucleus raphe magnus + nucleus raphe obscurus + nucleus raphe pontis + postpyramidal nucleus + inferior central nucleus, etc.).

However, to reduce the risk of suggesting homologies where they may not exist, we describe here only the broad categories of labeled cell groups in order to identify only the unequivocal commonalities among the animals.

### Labeled cell groups common to all 22 species

For convenience, the results are divided into two parts: first a description of the labeled cell groups found in each of the 22 species (that is, nuclei which were invariably labeled) and then a description of cell groups labeled in only a few species.

The first of these two parts is further subdivided by the level of the labeled cell groups: hindbrain, midbrain, and then forebrain subdivided still further into diencephalon and telencephalon. Within each of these categories, the labeled cell groups are listed roughly in order of the abundance or percentage of labeled neurons within them.

**Hindbrain.** Within the medulla and pons, 14 separate cell groups, nine major and five minor, were labeled in every species.

**Bulbar reticular formation.** By far the largest numbers of neurons labeled in the brainstem were found in the bulbar tegmentum. Labeled neurons were observed at all levels of the medulla in every individual—in the gigantocellular tegmental field, the parvocellular tegmental field, the lateral reticular nucleus, and in clusters surrounding the facial nucleus. These neurons were sufficiently filled with HRP reaction product that even fine dendritic branches could be seen extending from the somata. Occasionally, the axons from gigantocellular neurons could be seen coursing dorsally and medially to cross the midline (Figs. 4, 11, 14).

Most labeled gigantocellular neurons were found on the side contralateral to the hemisection while labeled cells in the parvocellular nuclei were found primarily ipsilaterally. The other labeled cell groups showed no clear asymmetry.

In addition to the cell groups in which labeled neurons were found in great abundance or in large percentage, a few labeled neurons were invariably seen in the periolivary nuclei of the superior olivary complex, especially between the lateral hilus of the lateral superior olive and the facial nerve (Fig. 18).

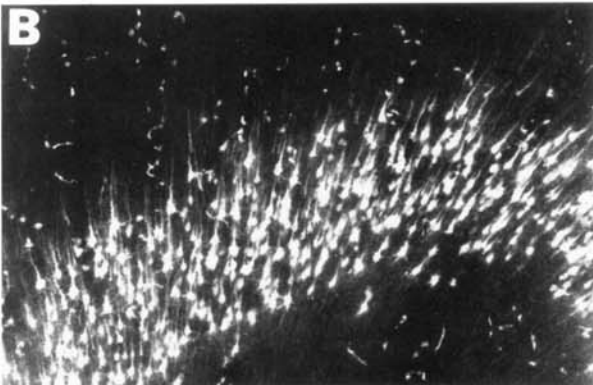
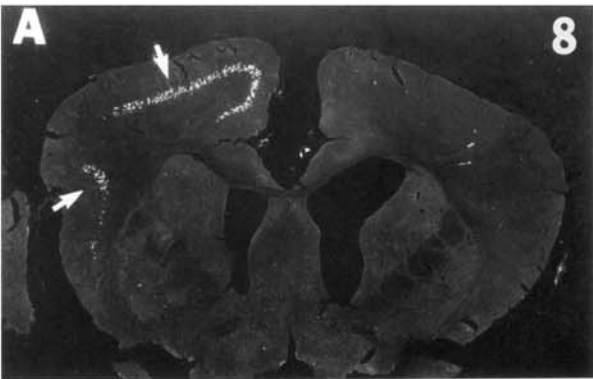
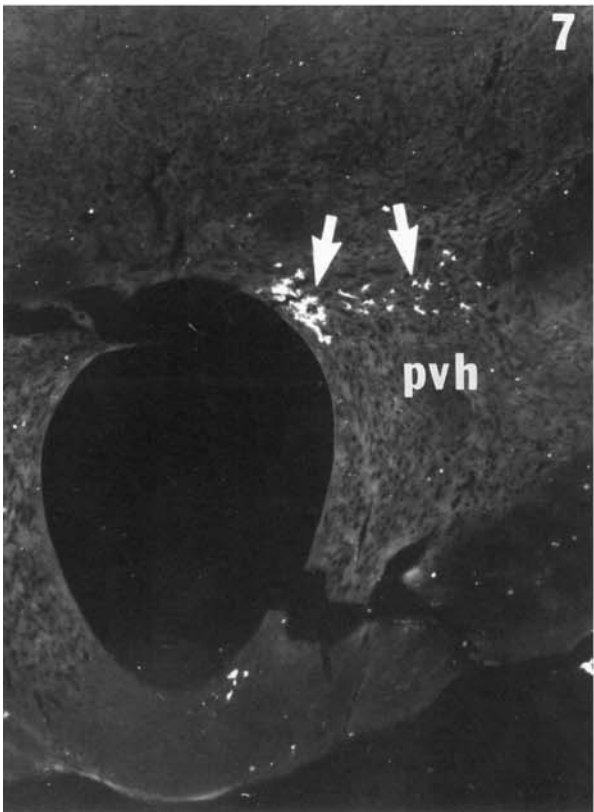
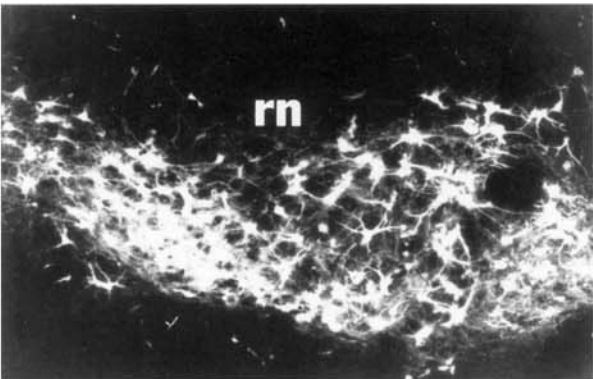
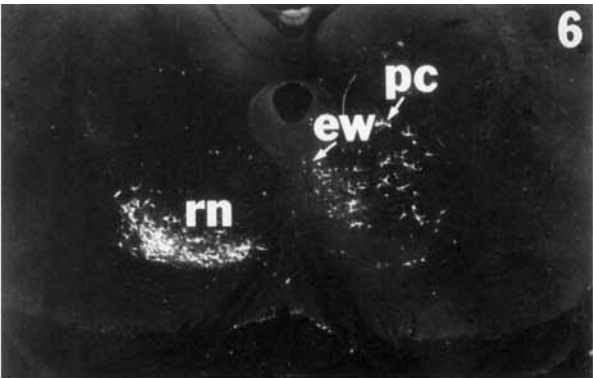
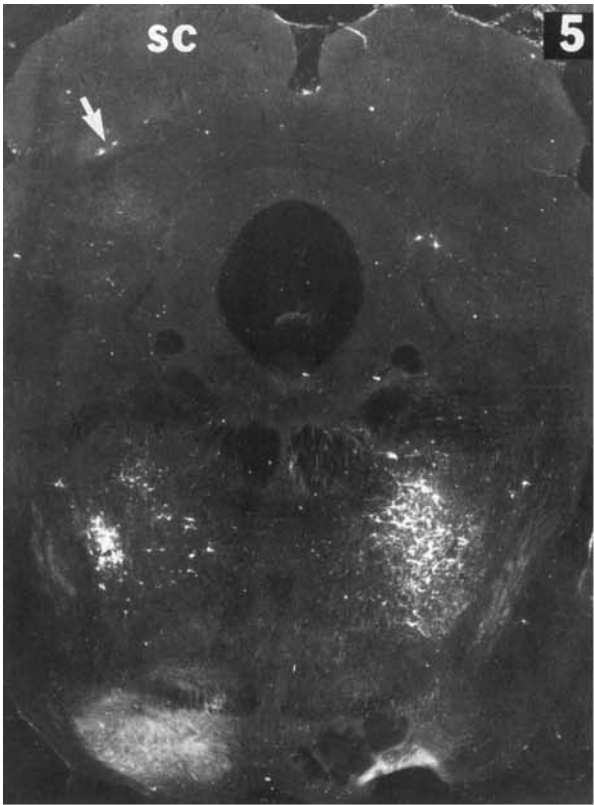
**Pontine reticular formation.** The heavy labeling of tegmental cell groups continued into the pons. A strict asymmetry of the labeled cells in the pontine reticular formation

Fig. 1. A: Dorsal view of the brain and upper cervical spinal cord of the bushbaby (*Galago senegalensis*). Arrowhead points to hemisection at the C1-C2 junction. HRP was applied to the cut fibers and the animal was allowed to survive for 72 hours. This figure, along with Figures 2–8, show the degree of HRP labeling throughout the neuraxis of a representative case. B: Low-power photograph (in negative) of section through spinal cord processed for HRP with TMB as the chromagen. This section was approximately 1.5 mm caudal to the hemisection. Note labeling primarily confined to right side of spinal cord. In the figures that follow, low-power ( $\times 20$  or less) photos were produced by projecting the actual tissue directly onto photographic paper and thus they appear in negative. The photos are arranged so that the hemisection always appears on the right side of the cord.

Fig. 2. HRP-labeled neurons in the spinal cord (of bushbaby). A: Level of cervical enlargement approximately 10 cm below right hemisection. Note large number of labeled neurons in both ipsilateral and contralateral spinal gray. B: Level of lumbar enlargement. Large number of labeled neurons at long distances from the hemisection provide added assurance that the technique marked most if not all neurons whose axons were severed by the cut.

Fig. 3. HRP labeling in two sections at the level of the pyramidal decussation (of bushbaby). Note intense labeling of fibers (white arrowheads) crossing the midline to form the medullary pyramid.

Fig. 4. Two sections at the level of the hindbrain (of bushbaby). A: Note large number of HRP-labeled somata in raphe nucleus (r), vestibular nuclei (lv), and gigantocellular neurons of the reticular formation (rf). Also note strict asymmetry of fiber labeling in medullary pyramid (pyr) indicating confinement of hemisection to right side of spinal cord. B: A higher-power section through the same level as shown in A, illustrating typical labeling of HRP-filled neurons. Note labeled axons (indicated by arrows) originating from gigantocellular reticular neurons.



Figures 5–8

was clearly evident in each species (Figs. 5, 20). Labeled cells in the ventrolateral part, or paralemniscal tegmental field, were always most abundant on the contralateral side while labeled cells in the central tegmental region were most abundant on the ipsilateral side.

Many labeled neurons were invariably observed just medial to the fibers of the lateral lemniscus with a few interspersed in the most medial of the lemniscal fibers. Labeled dendrites emanating from the paralemniscal neurons often could be seen coursing into the lemniscus itself.

**Vestibular complex.** The largest number of labeled vestibulospinal neurons were found in the dorsal and ventral divisions of the lateral vestibular nucleus (i.e., Dieter's nucleus). The somata of these cells were always among the largest of labeled neurons observed anywhere, especially in the dorsal division (Figs. 4, 14, 15). Most if not all of the neurons in this nucleus were labeled in every case, especially on the ipsilateral side. However, numerous labeled cells were also seen in the inferior, medial, and superior vestibular nuclei.

**Descending or spinal nucleus of V.** In every case, a large number of neurons were labeled in each subdivision (caudalis, interpolaris, oralis) along the rostrocaudal extent of the spinal nucleus of the trigeminal nerve. These neurons were always found bilaterally although there were slightly more labeled neurons ipsilateral to the hemisection. The largest number of neurons was found in the pars oralis, in which they tended to cluster near the most ventral aspect of the caudal portion of the nucleus. In more rostral sections through pars oralis, they invariably appeared in regular staggered arrangements, with their dendrites oriented dorsolaterally to ventromedially. In every case, labeled cells were also seen clustered around, but not within, the motor nucleus of V (Figs. 14, 16).

**Raphe complex.** Many neurons were labeled in the raphe nuclei in every case, primarily in nucleus raphe magnus, but also in nucleus raphe pallidus and nucleus raphe obscurus (Figs. 4, 11, 14). Occasionally, labeled neurons were found ventrally to nucleus raphe pallidus embedded within the fibers of the pyramidal tract.

**Deep cerebellar nuclei.** A large number of labeled neurons were always seen approximately midway through the rostrocaudal extent of the fastigial and interpositus nuclei of the cerebellum, primarily on the contralateral side. Most

labeled cells were located in the most medial portions of the fastigial nucleus embedded in fibers coursing between the fastigial and interpositus nuclei. In addition, a smaller number of labeled cells were seen in every case within the interpositus nucleus and along the border between the interpositus and dentate nuclei (Fig. 13).

**Dorsal column nuclei.** Although largely obscured by heavy anterograde labeling, labeled neurons in the dorsal column nuclei were always most numerous in the cuneate with fewer in the gracile. Labeled somata were seen primarily along the ventral border of the ipsilateral nucleus (Fig. 10).

**Nucleus ambiguus.** Labeled cells in the contralateral nucleus ambiguus always formed a well-defined cluster of relatively large, well-filled elongated multipolar neurons at the level of the inferior olivary nucleus. Labeled dendrites of these neurons were typically oriented from dorso-medial to ventrolateral (Figs. 9, 10).

**Nucleus of the solitary tract.** Labeled cells were always found in both the medial and lateral nucleus solitarius. Most were lateral and ventral to the tract (Fig. 12).

**Nucleus intercalatus.** Several labeled neurons were invariably observed in the nucleus intercalatus between the dorsal motor nucleus of X and the hypoglossal nucleus.

**Nucleus intertrigeminalis.** Labeled cells were found scattered along the medial borders of the principal sensory nucleus of V in the nucleus intertrigeminalis (Taber, '61). Though they were always small and few in number they appeared in every case (Fig. 15).

**Nucleus of Kolliker-Fuse.** A few labeled neurons were invariably observed in the nucleus of Kolliker-Fuse. Those that were labeled were small- to medium-sized multipolar or elongated cells. A few other labeled neurons were observed scattered throughout the parabrachial region (Fig. 19).

**Hypoglossal nucleus.** While a few labeled neurons were observed along the ventrolateral edge of the hypoglossal nucleus in every case, few of the large (and presumably, motor) neurons were labeled.

**Dorsal motor nucleus of X.** In every case, a few labeled neurons were observed at the lateral and ventral borders of the dorsal motor nucleus of X. They were seen almost exclusively near the obex, caudal to the pyramidal decussation.

**Midbrain.** Within the midbrain, ten cell groups, six major and four minor, were found labeled in each animal.

**Red nucleus.** Almost every cell in the contralateral magnocellular division of the red nucleus was vividly labeled in every case (Figs. 6, 9, 21, 23). In animals in which a distinct border around the parvocellular division could be reliably discerned, labeled cells were also observed surrounding that division but not within it.

**Interstitial nucleus of Cajal.** The labeled cells in this nucleus always formed a relatively compact group located in the rostral part of the midbrain at the level of the oculomotor nucleus and ventrolateral to the central gray. While labeled cells were found bilaterally, they were generally most abundant on the ipsilateral side (Figs. 6, 24).

**Coerulean complex.** Labeled cells were found in the locus coeruleus and subcoeruleus in each case. Labeled cells in the locus coeruleus were always most abundant on the ipsilateral side, primarily in the ventral part of the nucleus. Along the ventral edge of the locus coeruleus, labeled cells continued into the nucleus subcoeruleus (Fig. 19).

Fig. 5. Low-power section through the level of the superior colliculus (of bushbaby). The sparse labeling in the contralateral superior colliculus was found in most species studied (but see Fig. 22). Note strict asymmetry of tegmental cell groups, with labeled cells in the paralemniscal tegmental field more abundant on the contralateral side, and the cells in the central tegmental region more abundant on the ipsilateral side.

Fig. 6. Two different sections through the level of the red nucleus (rn) (in the bushbaby). While only a few rubrospinal neurons were labeled in the ipsilateral (right) red nucleus, virtually every neuron in the contralateral magnocellular division of the red nucleus was densely filled by HRP reaction product (also see Fig. 9). In addition to the red nucleus, note labeling of the nucleus of the posterior commissure (pc) and the nucleus of Edinger-Westphal (EW).

Fig. 7. Level of hypothalamus (of the bushbaby) showing labeled neurons in paraventricular nucleus of the hypothalamus (pvh).

Fig. 8. A: Typical section through the parietal cortex (of the bushbaby). Hundreds of cortical neurons per section were labeled. Note the strict contralaterality of the labeled cortical cells in this case and their restriction to layer V. B: Higher-power photo of typical section through region of highest numerical density of labeled neurons in the rostral cortex.

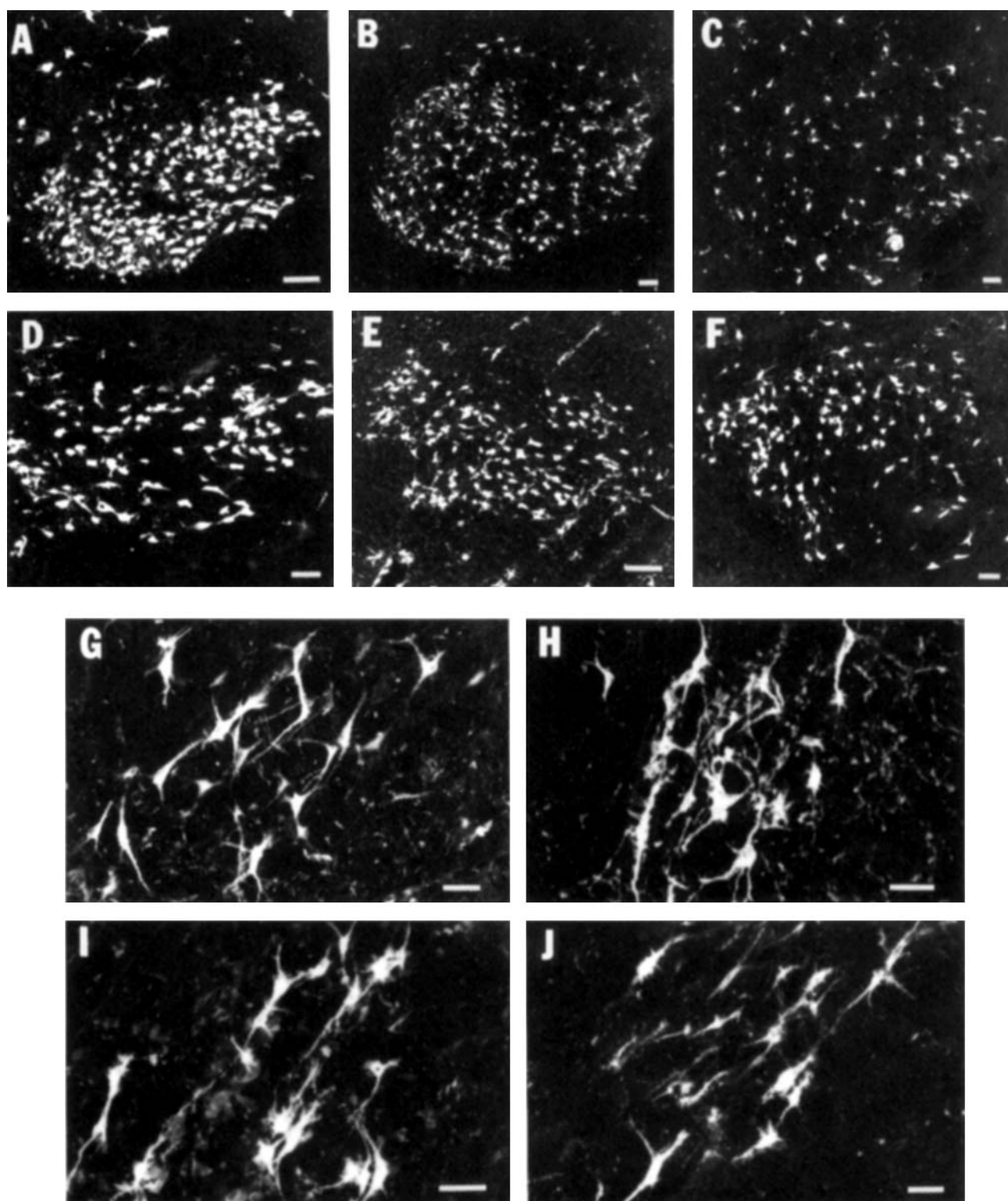


Fig. 9. HRP-labeled neurons in the magnocellular division of the red nucleus of six species and the nucleus ambiguus in four species. Top (red nucleus): In each individual examined in this study, virtually every neuron in the contralateral magnocellular division of red nucleus was densely filled by HRP reaction product: Cells in parvicellular division were unlabeled. A: Ground squirrel, B: Cat, C: Green monkey, D: Marmoset, E: West African hedgehog, F: Squirrel monkey. Bottom (n. ambiguus): In each individual,

many labeled neurons were observed in the contralateral nucleus ambiguus at the level of the inferior olivary nucleus. G–J: Labeled somato and primary dendrites characteristic of the nucleus (Taber, '61). Dendritic fields were primarily oriented dorsomedial to ventrolateral in each case. G: European hedgehog, H: Short-tailed opossum, I: Bushbaby, J: Raccoon. Calibration bars: red nucleus = 100  $\mu$ m, n. ambiguus = 50  $\mu$ m.

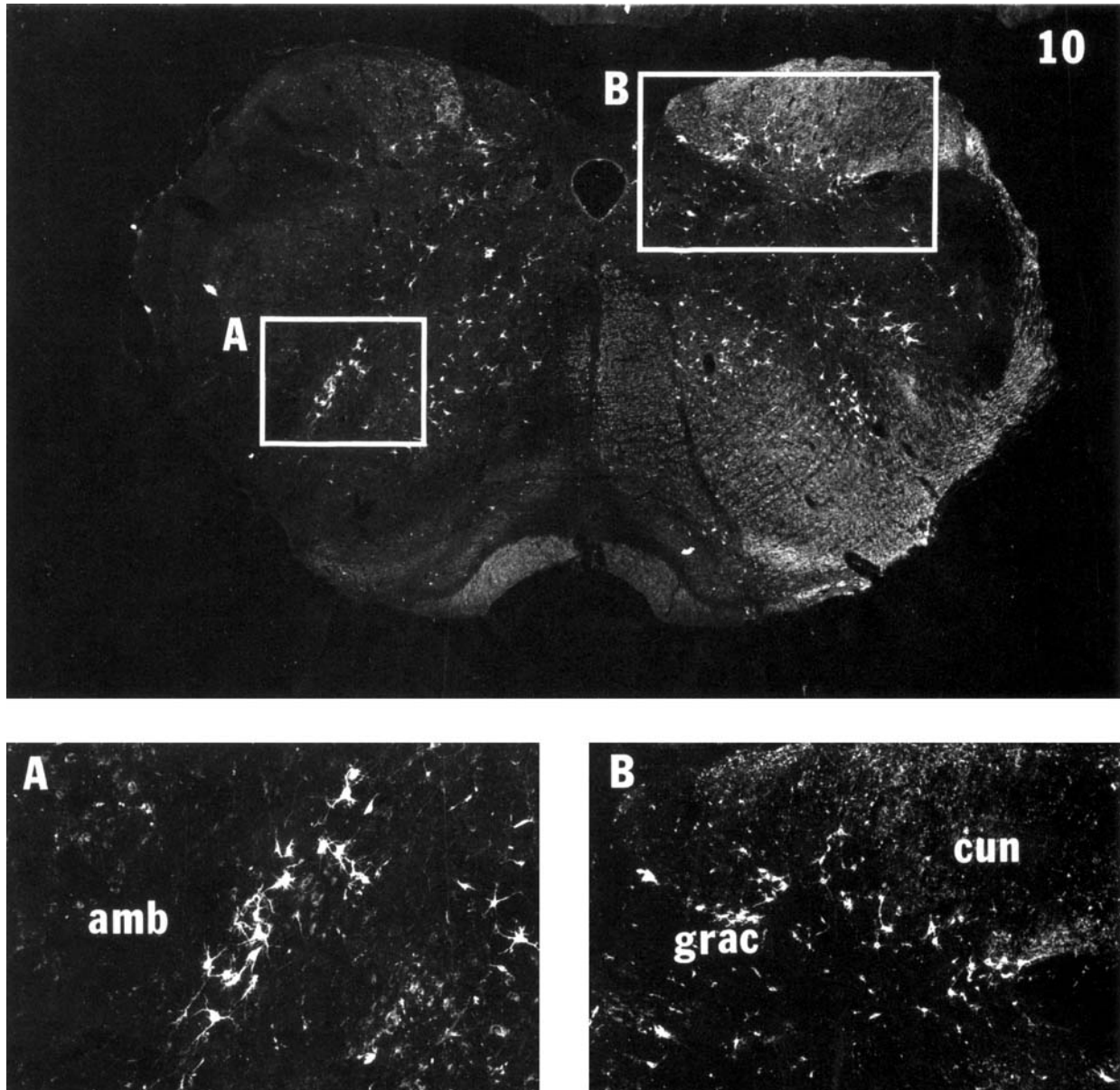


Fig. 10. Low-power section and higher-power enlargements through the lower medulla (of the ground squirrel). A: Labeled neurons in the nucleus

ambiguus (amb): compare with Figure 9. B: Labeled neurons in and around the dorsal column nuclei (grac = gracile nucleus; cun = cuneate nucleus).

**Nucleus of Darkschewitsch.** A moderate number of labeled neurons were always observed in the nucleus of Darkschewitsch embedded in the ventrolateral part of the central grey and dorsomedial to the interstitial nucleus of Cajal.

**Cuneiform nucleus.** Numerous labeled cells were always observed scattered throughout the rostral-caudal extent of the cuneiform nucleus ventrolateral to central gray. The vast majority of these cells were located ipsilateral to the hemisection (Fig. 21).

**Central gray.** Labeled neurons were scattered throughout the ventrolateral portion of the mesencephalic central gray in each case. They were located primarily ipsilateral to the hemisection.

**Superior colliculus.** Labeled neurons were observed in

both the intermediate and deep layers of the superior colliculus in every case. These neurons were large or medium sized and most were found laterally, primarily on the contralateral side (Figs. 5, 21–23). A wide and taxon-related variation in the number of labeled tectal neurons was observed among the species studied. This variation in the tectospinal tract and its correlates are the topic of a separate report.

**Edinger-Westphal nucleus.** In every case, several labeled neurons were observed in the Edinger-Westphal nucleus, especially near the midline rostral to the nucleus of n.III (Figs. 6, 23). Labeled cells continued anteriorly along the midline to the mesencephalic-diencephalic border. Most of these midline cells were elongated medium-sized neurons.

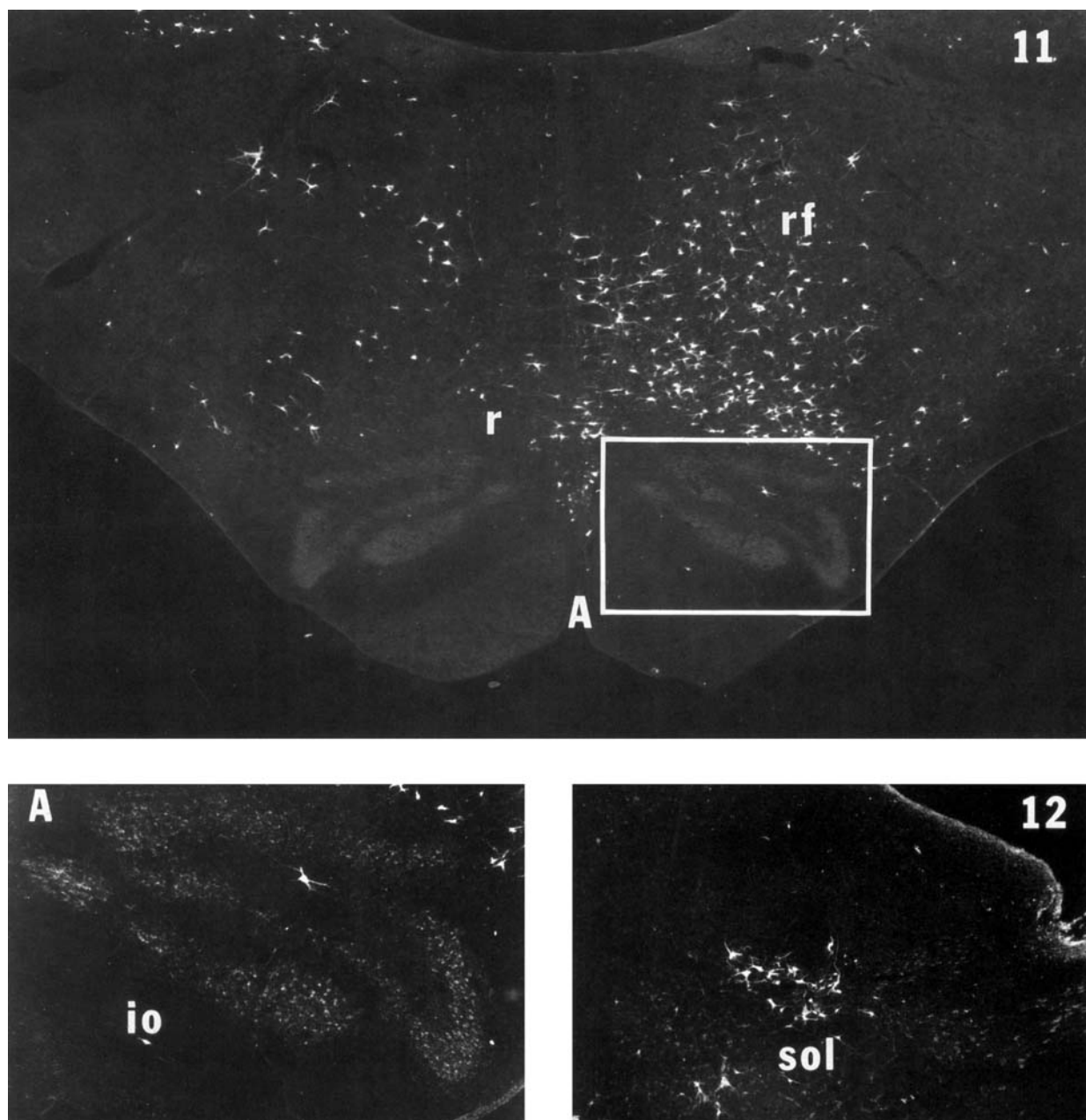


Fig. 11. Low-power section and higher-power enlargement (A) through level of inferior olive (io; of cat). Though few in number, labeled cells were occasionally seen between the cell groups of the inferior olives, with dendrites extending into the main cell groups.

*Nucleus of the posterior commissure.* A few small labeled neurons were invariably found in the nucleus of the posterior commissure just ventrolateral to the commissural fibers (Figs. 6, 24).

*Mesencephalic nucleus of V.* Very few neurons were labeled in the mesencephalic nucleus of V—usually less than ten per individual and almost exclusively on the ipsilateral side (Fig. 23). Those that were labeled were typically large

Fig. 12. Labeled cells in the nucleus of the solitary tract (of the cat). In each species, most of the HRP-labeled neurons were located ventrolateral to the tract.

and spherical in shape with characteristically short, thin dendrites. None of these neurons was vividly labeled and the HRP granules were usually small and somewhat dusty in appearance. Because the labeling of these somata was consistently different from those in every other labeled cell group, we suspect (but cannot prove) that their axons may end somewhat short of the spinal cord hemisection. We return to this question below.

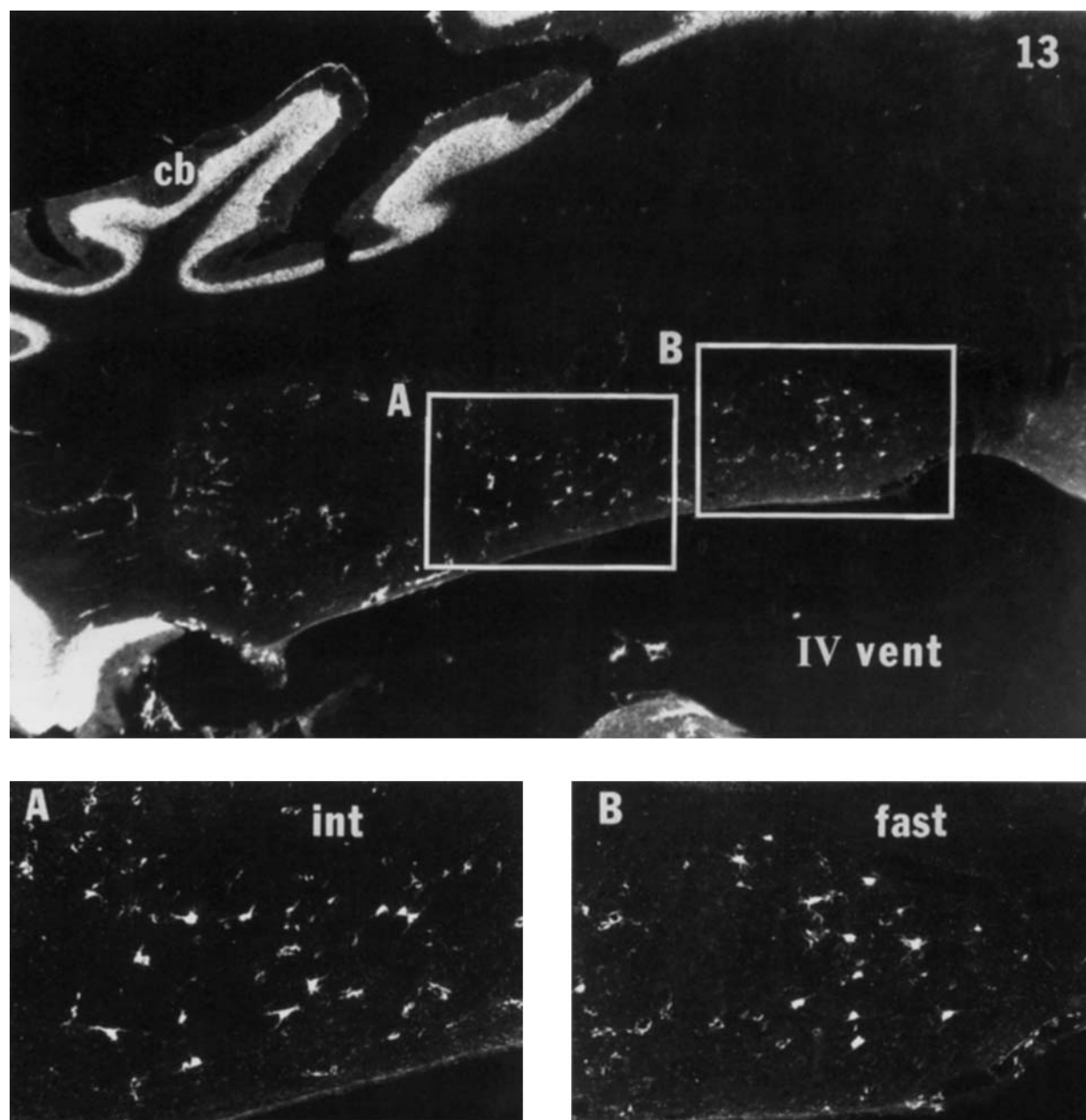


Fig. 13. Labeled cells in deep cerebellar nuclei (of marmoset). Note that several labeled cells are evident in both the contralateral nucleus interpositus and fastigial nucleus. A: Nucleus interpositus (int). B: Fastigial nucleus

(fast). (IV vent = IV ventricle). This section was counterstained with cresyl violet.

**Diencephalic cell groups.** Within the diencephalon, two labeled cell groups, one major and one minor, were found in every species.

**Hypothalamus.** Many labeled neurons were observed among the various nuclei of the hypothalamus of each species studied. They were always most abundant in the ipsilateral dorsal paraventricular and periventricular nuclei. Fewer labeled cells were observed in the lateral, ventromedial, and posterior hypothalamus in each case (Figs. 7, 27).

**Zona incerta.** Labeled cells in zona incerta were typically few in number but invariably present. The largest

number of labeled cells in zona incerta was found in rat. These cells were always situated in the most medial portion of the nucleus. Some of the most medial of these are probably located in the Fields of Forel (Fig. 25).

**Telencephalic cell groups.** Only one cell group (cortico-spinal) was found labeled in the telencephalon of every species.

A very large number of labeled neurons appeared in layer V (and only in layer V) of the cerebral neocortex in each of the species studied (Figs. 8, 27). With few exceptions among the species, more labeled neurons were observed in the cerebral cortex than in any other cell group. In most spe-

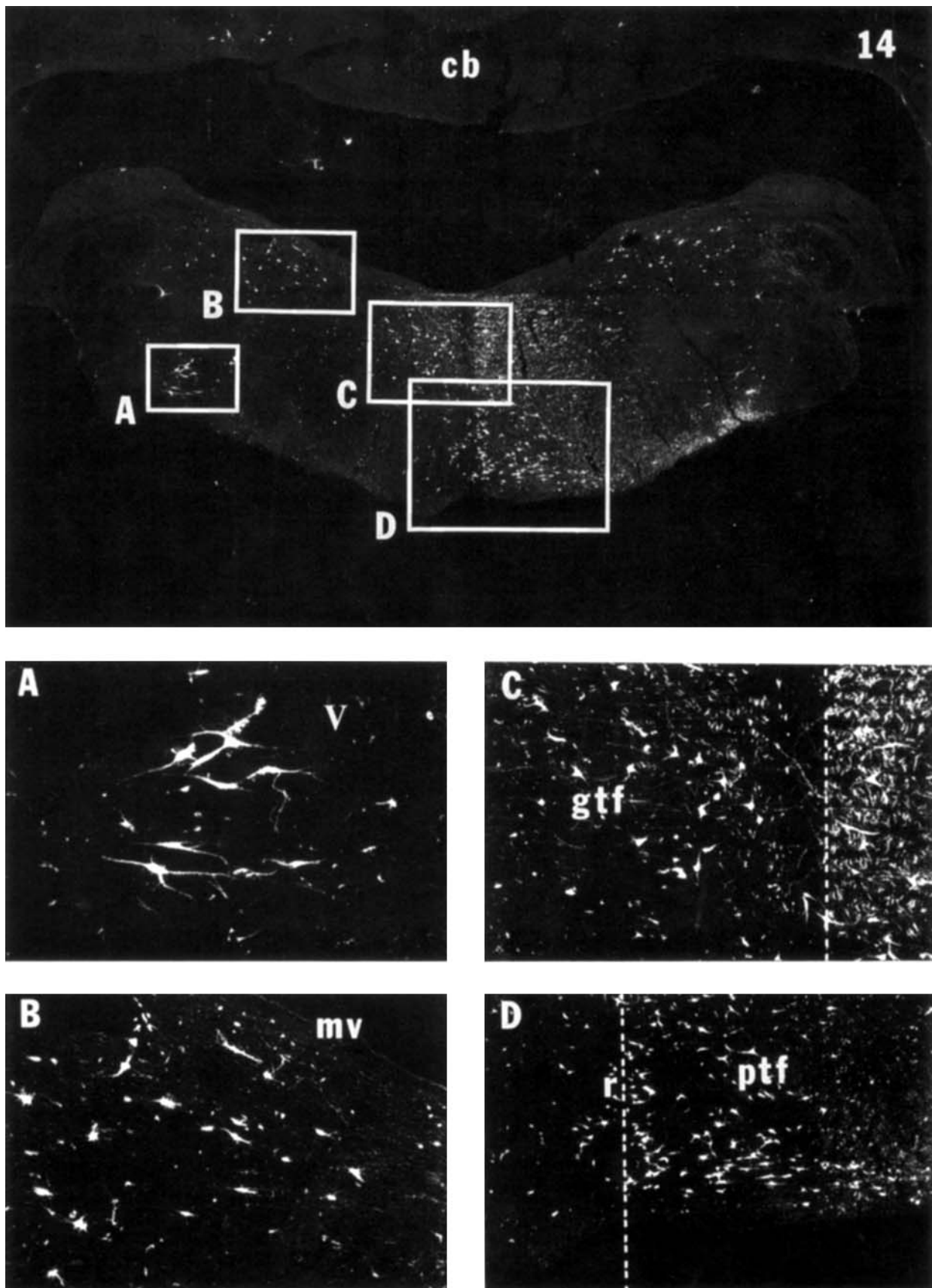


Fig. 14. Low-power section and higher-power enlargements through the level of the rostral medulla (of the hedgehog). A: Spinal V. B: Medial vestibular nuclei (mv). C: gigantocellular tegmental field (gtf). D: Parvocel-

lular tegmental field (ptf) and raphe nuclei (r). Vertical dashed line indicates midline.

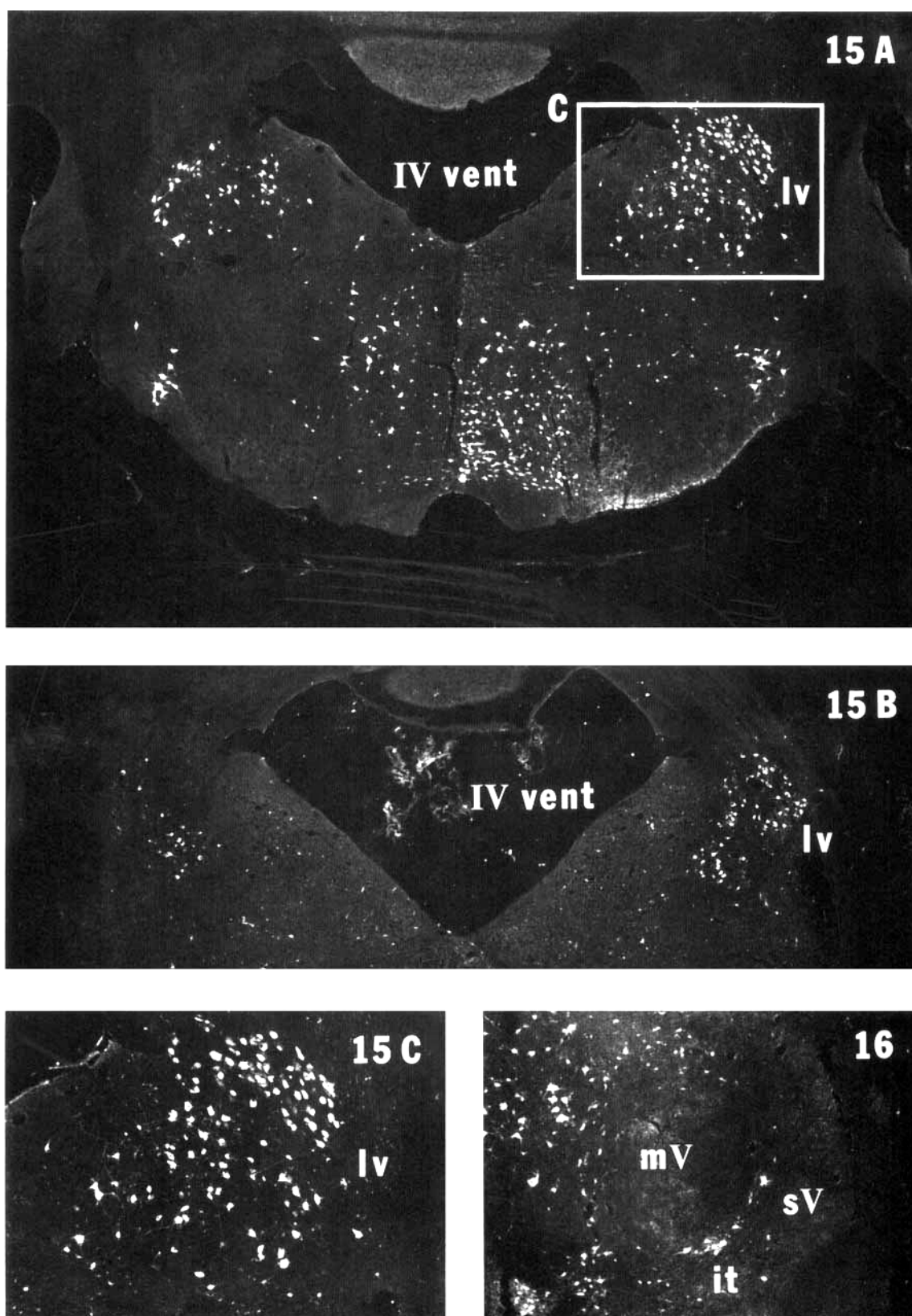
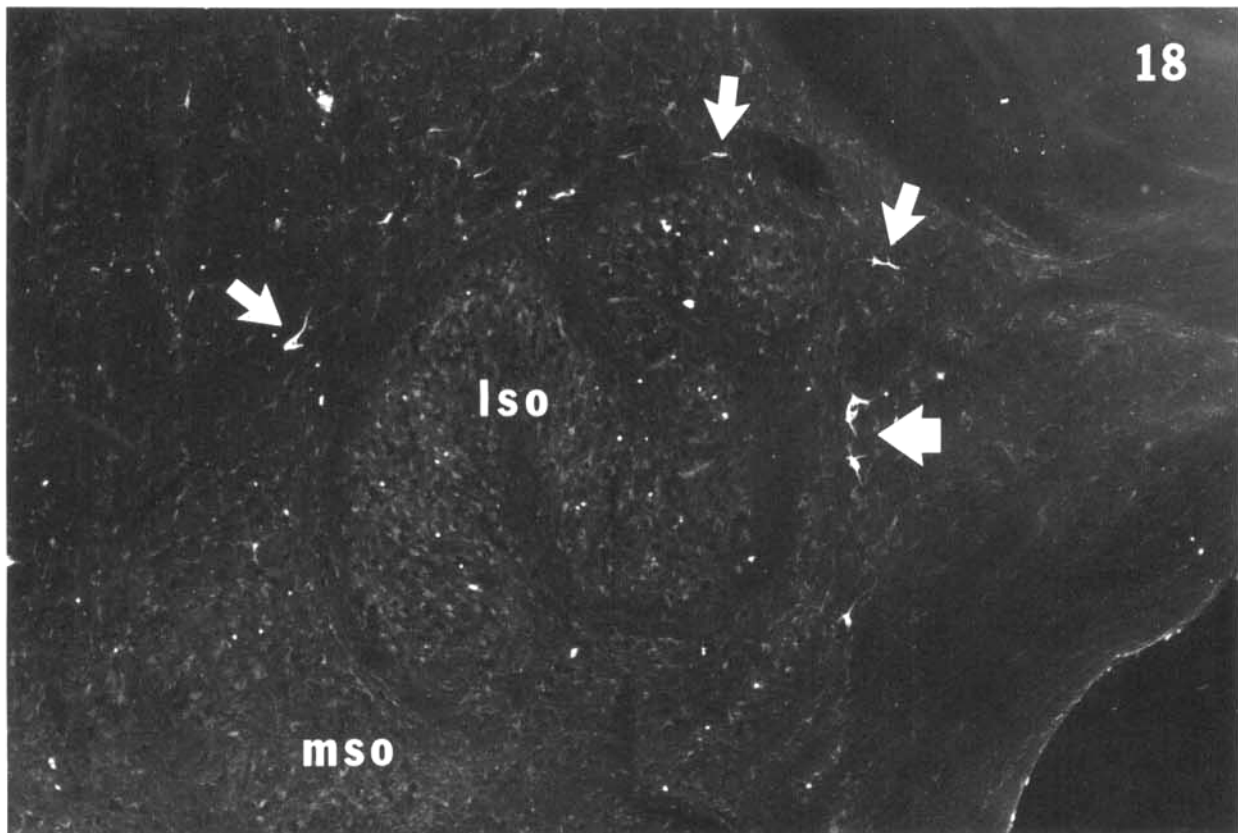
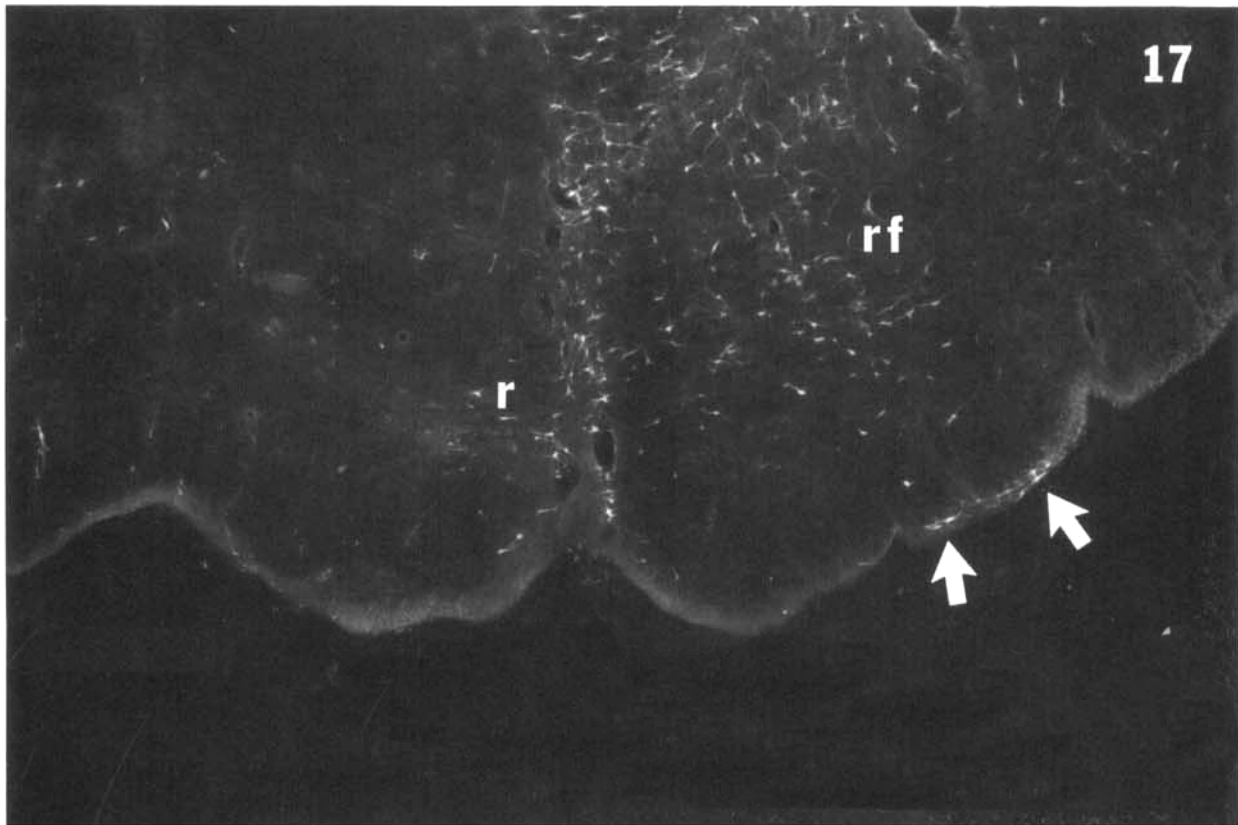


Fig. 15. Comparison of labeling in lateral vestibular nuclei (lv) in two different species (ground squirrel and green monkey). In each species, labeled neurons were most abundant in the ipsilateral lateral vestibular nucleus. A: Ground squirrel. B: Green monkey. C: Higher-power photo of labeled cells in the lateral vestibular nuclei (ground squirrel). The somata of these cells were always among the largest of labeled neurons observed.

Fig. 16. Labeled neurons surrounding the motor nucleus of V (of ground squirrel). While the large motor neurons were never labeled, several labeled neurons were seen between motor V and the principal sensory nucleus of V in each individual examined, the intertrigeminal nucleus (it; Taber, '61). (sv = sensory n. of V).



Figures 17 and 18

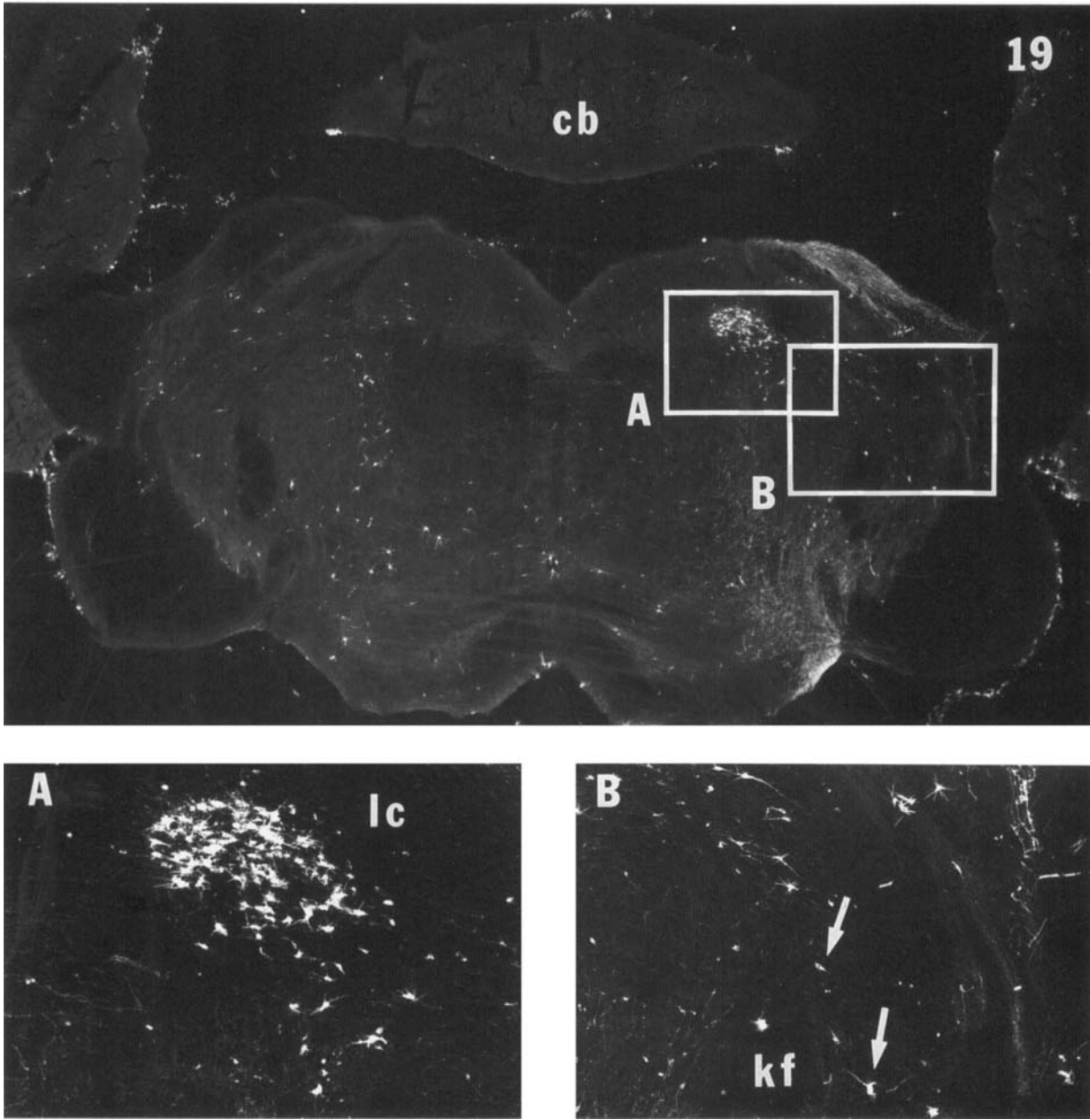


Fig. 19. Low-power section and higher-power enlargements through the level of the locus coeruleus (of short-tailed opossum). A: Locus coeruleus (lc). B: Nucleus of Kolliker-Fuse (kf) (arrows).

Fig. 17. Labeled cells ventral to the facial nerve nucleus of the armadillo (arrows). While labeled cells in the vicinity of the facial nerve nucleus were seen in every species, a distinct cluster of many neurons was seen only in armadillo (r = raphe; rf = reticular formation).

Fig. 18. Labeled cells surrounding the lateral superior olive (lso) (of rabbit). While a few labeled cells could be seen in at least four distinct cell groups (arrows), the majority of these presumed periolivary neurons were seen between the lateral hilus of the lateral superior olive and the emerging facial nerve. (mso = medial superior olive).

cies, the corticospinal neurons were located primarily in the contralateral hemisphere, with very few in the ipsilateral hemisphere. However, in the insectivore species (mole: *Scalopus aquaticus*; hedgehogs: *Erinaceus europaeus*, *E. al-biventris*, and *Hemiechinus auritus*; but not in the least shrew, *Sorex*, not included in the present study), the number of labeled cells in the ipsilateral hemisphere reliably exceeded the number in the contralateral hemisphere. Since corticospinal neurons are the focus of a specific report to follow, they will not be described further here.

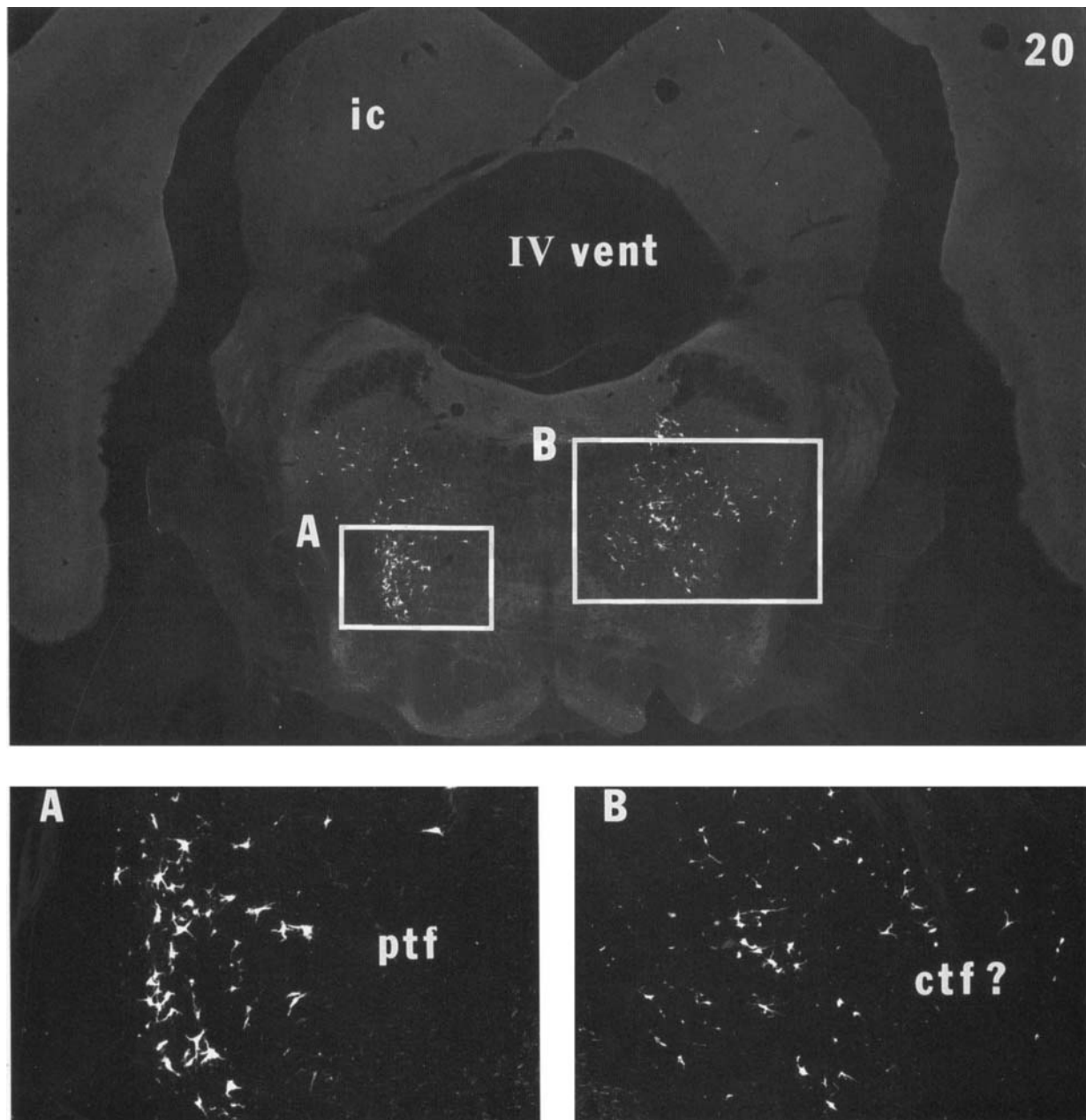


Fig. 20. Low-power section and higher-power enlargements through level of pontine tegmentum (of rat). A: Paralemniscal tegmental field (ptf). B: Central tegmental fields (ctf).

### Cell groups found labeled in only a few species

To this point, we have listed only those 27 generic cell-groups invariably found to contain labeled neurons in each of the 53 individuals of the 22 species. Here, we list five other cell groups found labeled in only a few species although found in each individual of those species.

**Amygdala.** In a few species, labeled neurons were found within the ipsilateral central nucleus of the amygdala (Fig. 11). These amygdalospinal neurons were found in one spe-

cies of opossum (*Monodelphis domestica*), in all three species of hedgehogs (*H. auritus*, *E. europæus*, *E. albiventris*), and in armadillo (*Dasypus novemcinctus*), and one labeled cell was found in tree squirrel (*Sciurus carolinensis*). These labeled neurons were few in number even among these species, reaching their maximum of 110 in opossum (*M. domestica*). They were found exclusively in the ipsilateral hemisphere.

Although specifically sought, no labeled neurons were found in the amygdala in any of the other species, although

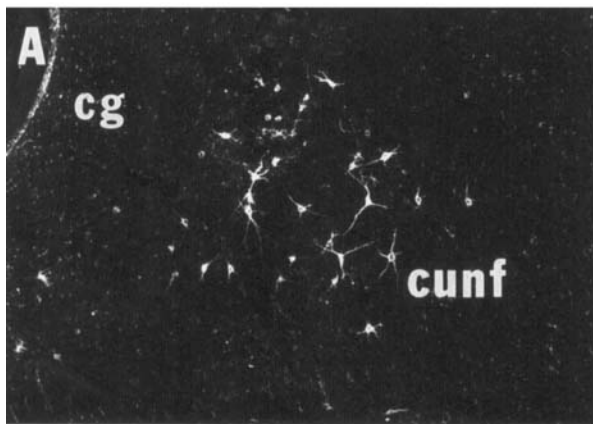
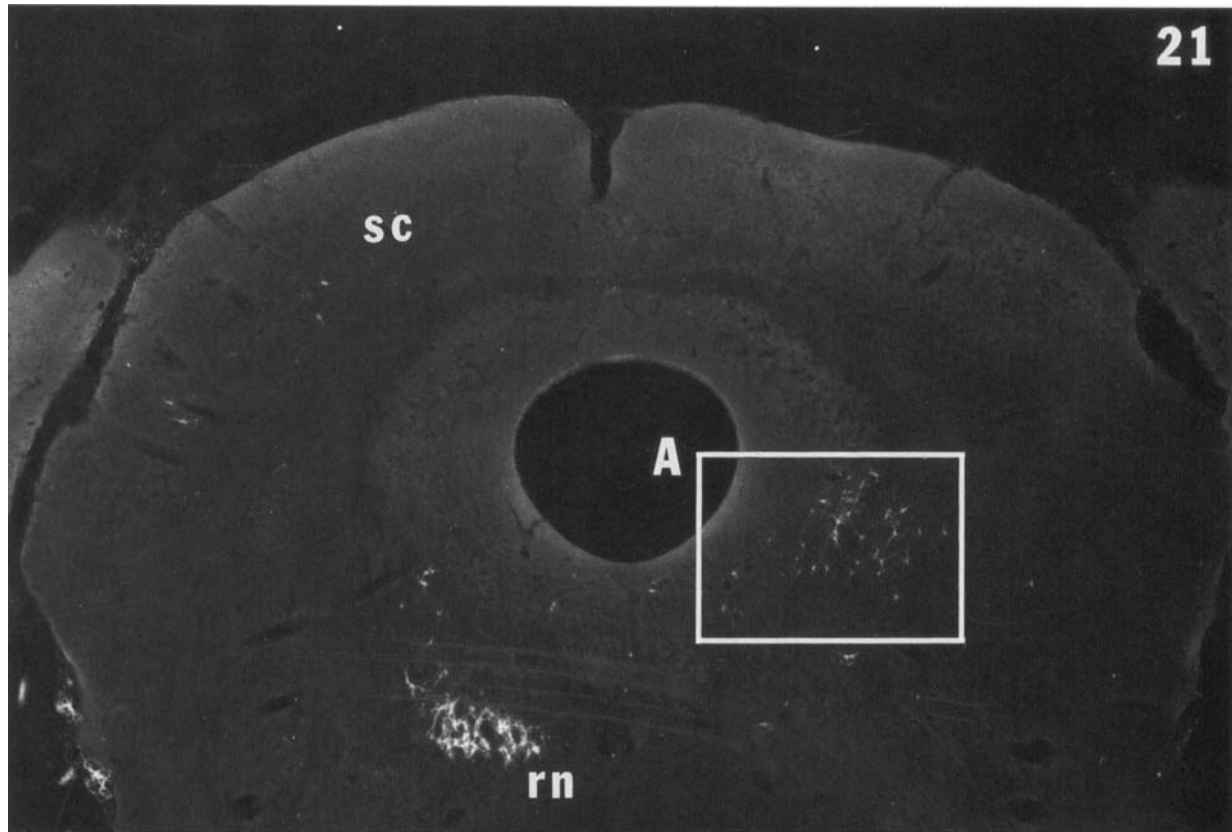


Fig. 21. Low-power section and higher-power enlargement (A) through level of midbrain (of the short-tailed opossum), showing labeled cells in lateral part of central gray (cg) and in the cuneiform nucleus (cunf).

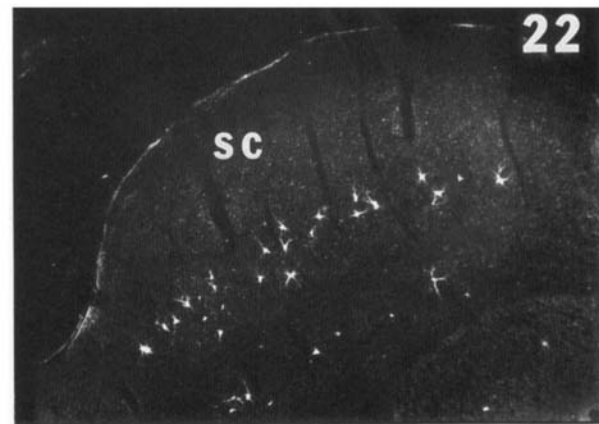


Fig. 22. Labeled cells in the intermediate and deep layers of the superior colliculus (sc) (of cat). While many labeled cells were seen in the superior colliculus of cat and raccoon, few were observed in most species examined (e.g., see Figs. 5, 21, 23.).

their relative paucity increased the likelihood of their being missed in the wider-spaced sets of sections through the larger brains. However, because of their small absolute number in the species in which they were found, and because they were observed in only six of the 22 species studied, amygdalospinal neurons might be considered to be only a minor (and possibly, variable) telencephalic source of spinal afferents. We return to this point below.

**Inferior olive.** A few labeled cells (only three to four neurons per individual) were found between the cell-rich layers of the inferior olive only in cat. The dendrites of

these cells were observed extending into the main cell groups (Fig. 11).

**Neurons ventral to the facial nucleus.** In armadillo alone, several labeled neurons were found in a cell group near the ventral surface of the brainstem beneath the facial nucleus (Fig. 17).

**Stria medullaris.** Only one group of cells in or near the thalamus was seen to contain labeled neurons in any of the species. These cells were embedded within the fibers of the stria medullaris (Travis, '82) only in the armadillo (*D. novemcinctus*) (Fig. 26).

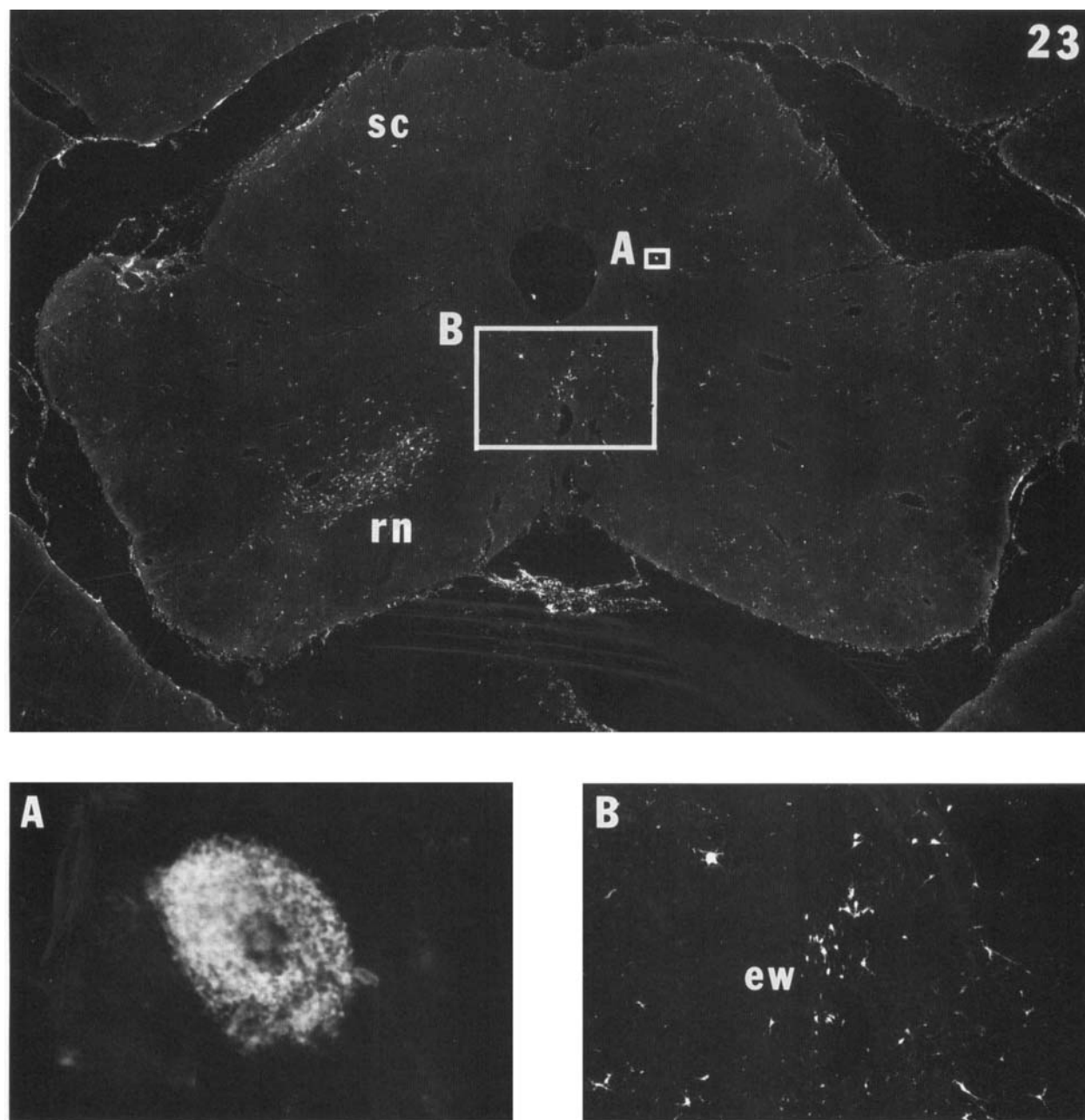


Fig. 23. Low-power section and higher-power enlargements through the level of red nucleus (of armadillo). A: Atypical labeling of cell in the ipsilat-

eral mesencephalic nucleus of V. B: Labeled cells in the nucleus of Edinger-Westphal (ew).

**Cells ventral to anterior commissure.** Finally, a few labeled cells were observed only in one species of hedgehog (*E. europæus*) and in one species of opossum (*M. domestica*) just ventral to the anterior commissure and exclusively on the ipsilateral side.

## DISCUSSION

### Possible sources of experimental error

In the present study, 27 distinct cell groups were found to be retrogradely labeled from the spinal cord in each of the 22 species. Short of technical error therefore, it is probably safe to conclude that these cell groups probably originate

axons descending directly to or beyond the C1-C2 level of the spinal cord. However, before discussing the implications of these results for a comparative analysis of descending pathways, several potential sources of uncontrolled variation warrant specific discussion. These possible sources of error can be conveniently divided into two types: false positives or overlabeling errors, in which cells might be labeled that, in fact, do not project to or past the C1-C2 junction; and underlabeling errors in which cells that do project to or past the C1-C2 junction somehow escaped detection. The extent to which each of these two types of error may have affected the results is discussed separately.

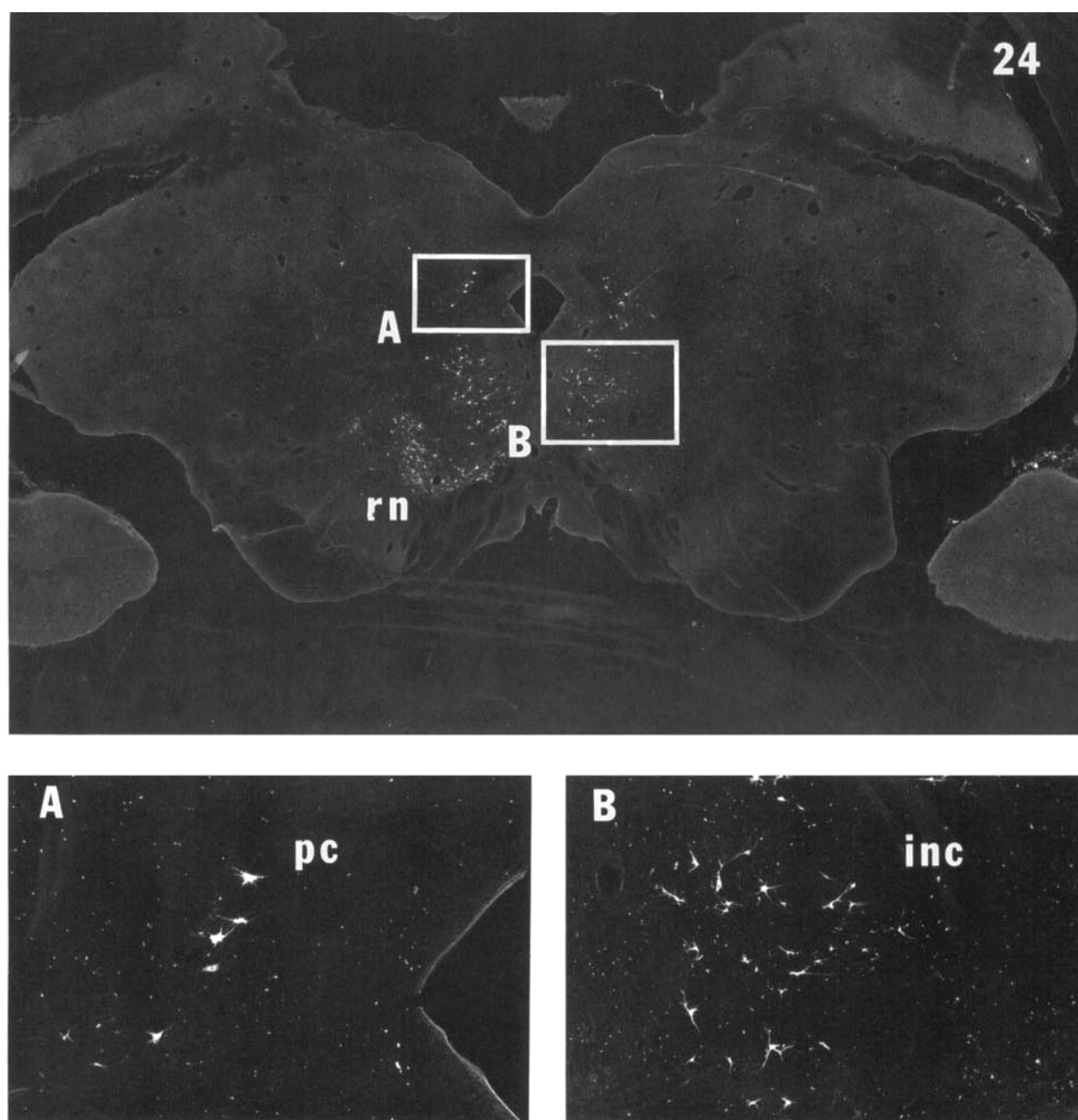


Fig. 24. Low-power section and higher-power enlargements through rostral midbrain (of cat). **A:** Labeled cells in the nucleus of the posterior commissure (pc). **B:** Labeled cells in the interstitial nucleus of Cajal (inc).

### Possible sources of overlabeling errors

We specifically considered five possible mechanisms by which cells which do not project to the spinal cord may have been erroneously labeled: vascular or CSF transport of the HRP, HRP diffusion into the medulla, endogenous peroxidase, and somata with dense terminal labeling on their surface masquerading as though retrogradely labeled. Each of these potential errors is discussed one at a time.

**Vascular transport of HRP.** Because of the somewhat radical surgical procedure employed here, and because of

the application of relatively high concentrations of peroxidase, it is possible that cells may have been labeled by vascular transport of the HRP to another site rather than, or in addition to, strictly axonal transport from the spinal cord. While this possibility can not be discounted lightly, no evidence of it could be found in the 53 cases reported here although it was specifically sought in every case.

To begin with, little commonality is found when comparing the present results with those obtained after intravenous injection of HRP. For example, many of the cell groups which are labeled by intravenous injection of HRP (such as

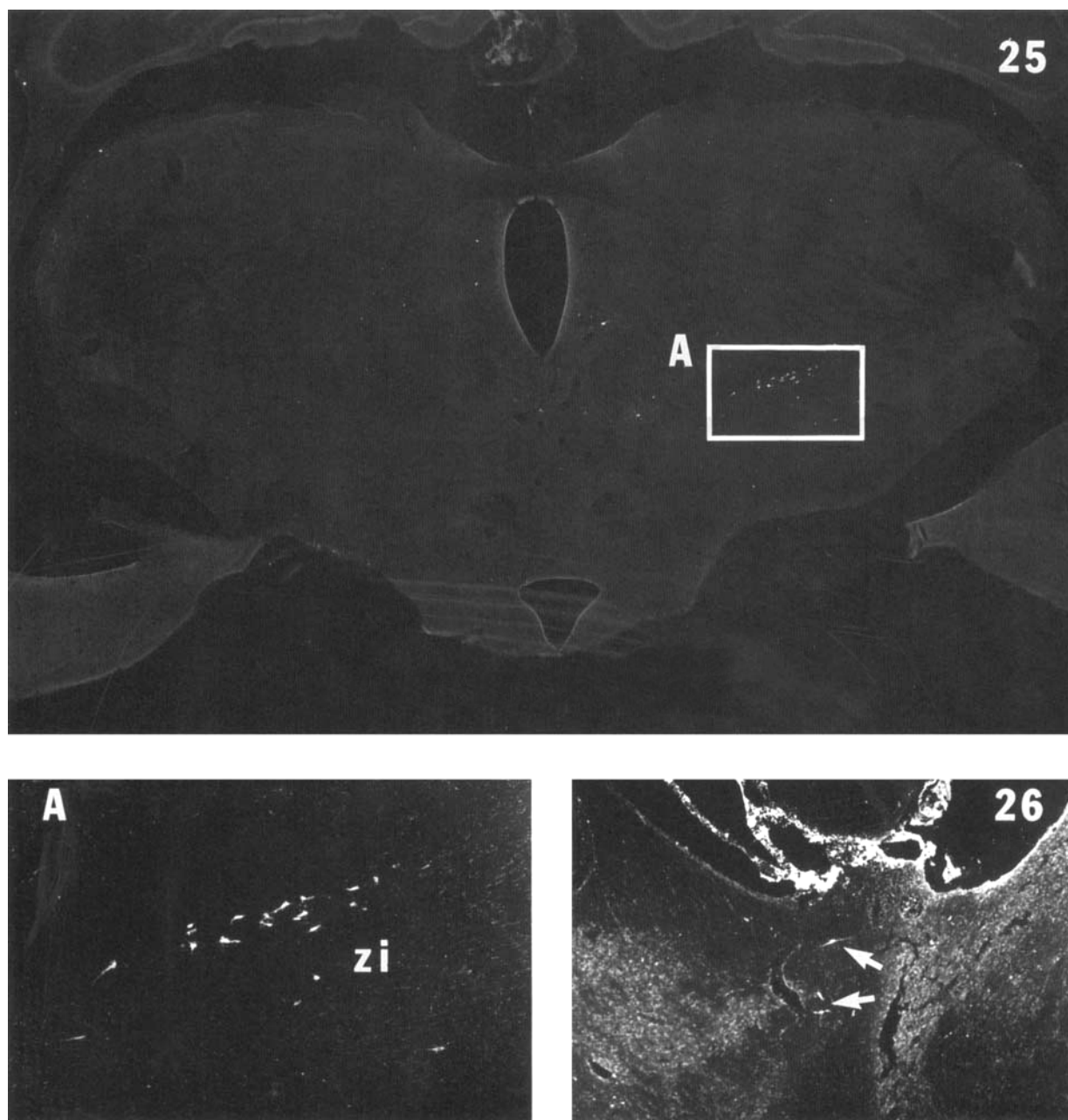


Fig. 25. Low-power section and higher-power enlargement (A) through the level of the zona incerta (zi) (of rat).

the motor nuclei of cranial nerves III–VII) were specifically not labeled in any of the cases in the present study (Broadwell and Brightman, '76). Also, after intravenous injections of HRP, the distribution of peroxidase-positive cells is almost always bilaterally symmetrical, while each cell group labeled in the present study, excepting very few, displayed clear asymmetry.

However, in one of the 94 cases studied (a squirrel monkey), HRP labeling was observed which was probably associated with vascular transport. In this case, bilaterally symmetrical labeling was seen in many neurons of n.III

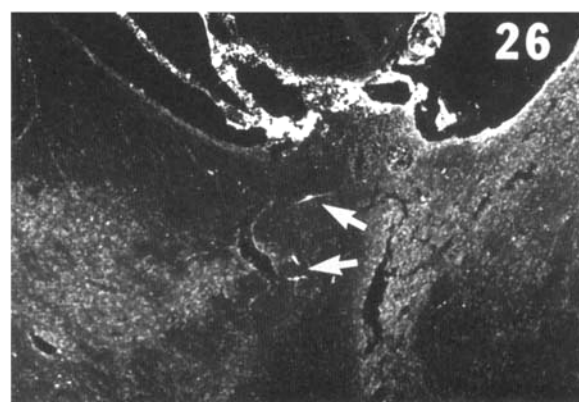


Fig. 26. Low-power section and higher-power enlargement through the level of the stria medullaris (of armadillo). This section was counterstained with cresyl violet.

and in the supraoptic nuclei. Since labeling in these nuclei has been demonstrated after intravenous administration of HRP, and since these same nuclei were not labeled in any of the 53 individuals upon which the present study is based, this case was judged to have been contaminated by intravenous transport and it was eliminated from the study. Though similar evidence was specifically sought in the other cases, no other individual yielded any evidence whatever of intravenous transport.

**CSF transport.** Despite precautions aimed at confining the diffusion of HRP from the site of application, it is

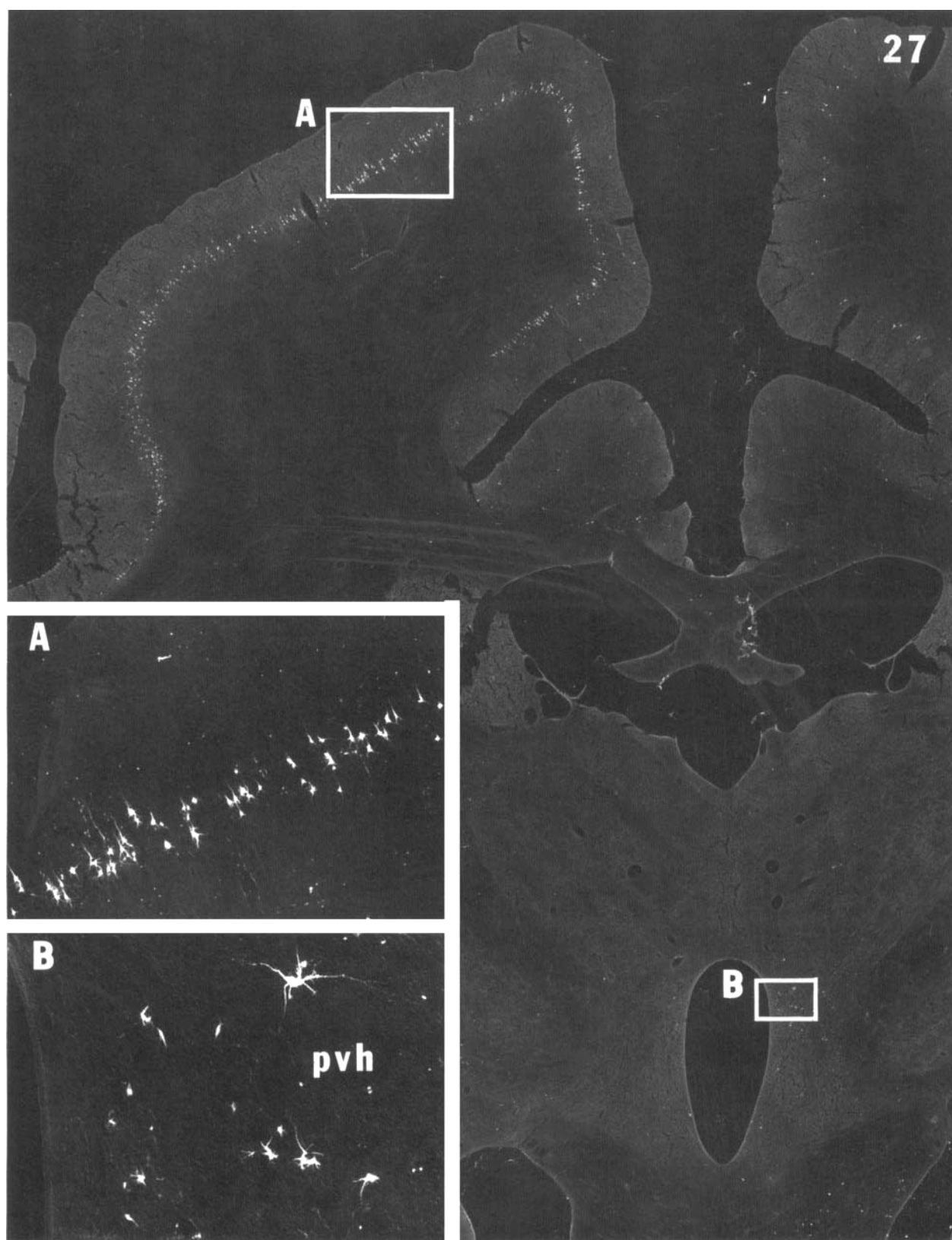


Fig. 27. Low-power section and higher-power enlargement through fore-brain (of macaque). **A:** Labeled cells in layer V of cerebral cortex. **B:** Labeled

cells in the ipsilateral dorsal paraventricular nucleus (pvh) of the hypothalamus.

possible that cells may have been labeled due to the transport of HRP to distant sites via cerebrospinal fluid. To be sure such uncontrolled distribution of the marker would be recognized and to assess its risk, control cases were prepared in which the dura was cut and retracted and HRP flakes were applied to the exposed but uncut cord. Despite a very large amount of HRP applied in these cases in an attempt to promote CSF transport, and despite overprocessing of the tissue with TMB, relatively few labeled cells were observed and those cells that did contain reaction product were only lightly labeled. This outcome stands in sharp contrast to the large number of densely labeled cells seen after applying HRP to the cut fibers of the spinal cord. Furthermore, the distribution of the few cells containing reaction product in the CSF control cases was always bilaterally symmetrical while bilaterally symmetrical labeling was not seen in cases in which the cord was hemisected and HRP applied to the cut fibers. Therefore, in the present study, overlabeling errors due to CSF transport of HRP are also unlikely.

**Extraaxonal diffusion.** Another mechanism by which overlabeling may have occurred is through rostral diffusion of the tracer beyond the medulla–spinal cord junction—that is, into lower medullary structures, particularly the dorsal column nuclei. As described before, this issue was a key factor in the experimental design. Indeed, the C1–C2 junction was chosen as the best site (over, say, C1 itself) specifically because any more rostral hemisection allowed obvious spread of the tracer into structures more rostral—that is, into the caudal medulla—particularly into structures in the medulla which are known to receive descending projections from still-more-rostral cell groups. For example, the dorsal column nuclei receive projections from sources which are also known to project to the spinal cord (Rustioni and Hayes, '81).

Because of the large numbers of fibers, both ascending and descending, which were labeled following HRP application to the hemisected spinal cord, it is not possible to describe accurately the limits of the spread of the tracer by examining TMB-treated tissue near the level of the cut. However, in tissue sets treated with the cobalt-DAB procedure instead, extracellular HRP reaction product appeared to be restricted entirely to the spinal cord. Thus, the present procedure probably effectively labeled cells whose axons extend as far as C2 and beyond (and hence were severed by the hemisection), and due to extra-axonal spread of the applied tracer, it probably also labeled endocytotically those axons which terminate at C1. However, because more rostral spreading of the marker was specifically sought and not found in each case reported here, the inadvertent diffusion of marker to bulbar terminals was probably negligible.

**Endogenous peroxidase.** We considered the possibility that some cells might be falsely labeled because of endogenous peroxidase activity (e.g., see Mensah and Finger, '75). For example, labeled cells of the mesencephalic nucleus of V contained reaction product which was atypically sparse and fine grained compared to other labeled cells in the same cases—a characteristic of intrinsic peroxidase activity. However, the reaction product in even these cells is almost certainly not due to endogenous peroxidase activity, since these cells were observed exclusively on the ipsilateral side. In general, the role of peroxidatic enzymes in TMB-treated tissue is now thought to be negligible, and we have no evidence that contraindicates that claim (Mesulam, '82). Therefore, errors due to endogenous labeling in the present cases seem unlikely.

**Orthograde vs. retrograde label.** Finally, a fifth source of potential overlabeling error is due to the possibility that a large population of labeled terminals on somata surfaces could have been mistaken for cytoplasmic granules. Because the density of reaction product in most labeled cells rendered the soma completely opaque, it was not usually possible to focus upon individual granules of reaction product within a plane passing through the interior of the soma. Although in some cells the unlabeled nucleus could be visualized within the completely filled cytoplasm, in most, even the nucleus was obscured. Therefore, the possibility that these densely labeled cells were covered with labeled afferents and not retrogradely labeled, too, cannot be entirely excluded.

However, several pieces of indirect evidence would seem to make the probability of mistaking orthograde for retrograde labeling quite low. To begin with, the present cut-and-fill technique with HRP as the tracer produced relatively weak orthograde labeling at levels above the lower medulla. Ascending pathways in the lower medulla, such as the medial lemniscus and spinocerebellar tracts, were intensely labeled orthogradely, and heavy perisomal orthograde labeling was also evident in the ipsilateral dorsal column nuclei and reticular formation. However, even this heavy labeling of terminals in these cell groups has probably not been mistaken for retrogradely labeled somata since the same somata were also labeled following use of other tracers (such as nuclear yellow) which are not good orthograde terminal markers.

Orthograde labeling in structures rostral to the medulla was sparse or absent entirely in the present materials, even in those structures which are known to receive a heavy projection from the spinal cord, such as the superior colliculus and the ventrobasal thalamus. Therefore, terminal labeling on somata rostral to the medulla was much too light to masquerade as retrograde labeling. For these reasons, the possibility of false positives due to dense and heavily labeled terminals on the surface of somata is also thought to be negligible.

In sum, at least five sources of potential false positives or overlabeling errors—those due either to vascular or cerebrospinal transport of HRP, rostral diffusion of the tracer, endogenous peroxidase activity, or mistaking orthogradely labeled terminals for retrogradely labeled somata—were specifically considered and in the 53 individual cases reported here judged to be negligible. Nevertheless, these potential errors cannot be ruled out entirely, and therefore the conclusions must be weakened accordingly.

### Possible sources of underlabeling errors

The second type of error, underlabeling errors, in which cells which do project to the spinal cord either escaped labeling or if labeled escaped detection, may also have occurred in at least two ways. First, the hemisection may not have been complete in some cases, and thus fibers close to the midline of the cord may not have been cut and exposed to the marker. Indeed, postfixation examination of the dorsal and ventral surfaces of the hemisected cord revealed that in three of the total of 94 cases, the section did spare a small portion of the ventromedian cord. These cases were among those discarded from the study. However, in each of the 53 cases reported here, histological examination showed that the cord section extended beyond the midline and that the entire cut surface of the cord had been exposed to the HRP.

The second mechanism by which underlabeling errors may have occurred is through HRP phobias, including phobias either specific to a particular tract or specific to a particular species or even specific to a particular tract in a particular species. The possibility of some of these HRP phobias can be ruled out due to the constancy and reliability of the HRP procedure within and among species. That is, an HRP phobia specific to particular individuals or to particular species can be ruled out since there was no individual case, and hence no species, whose axons did not transport HRP in at least the 27 tracts listed above.

Likewise, an HRP phobia specific to some tracts in some individuals can be ruled out since in those species in which more than one individual was examined, there was no qualitative difference among labeled cell groups. For the same reason, an HRP phobia specific to an interaction of species and tracts also can be ruled out for the 27 common tracts.

However, possible species-by-tract phobias may exist for those cell groups labeled in only a few species and not in the others. For example, some cells in the amygdala, others near the inferior olive, others near the stria medullaris, and still others ventral to the facial nerve nucleus were labeled in only a few species and not in others. It is important to note that the existence of possible species-by-tract phobias could have led to this result even if the particular tracts are indeed present in every species.

Another possible source of underlabeling error is one in which an HRP phobia may exist which is specific to one or more particular tracts in all species. The only evidence we can bring to bear on this possibility is that obtained by our trials of retrograde tracers other than HRP. For example, we used nuclear yellow, true blue, WGA-conjugated HRP, or iodine-labeled WGA as a marker in the preliminary experiments. Since we saw no cell group labeled by any of these markers that was not also labeled by our standard procedure, it can be concluded that if particular tracts are HRP phobic, they are probably phobic to each of these other markers as well. If indeed such tracts exist, their presence may evade discovery for some time to come. Therefore, though probably unlikely, an HRP tract phobia common to all species is not impossible, and for this reason, the present list of cell groups contributing descending pathways to the spinal cord cannot be considered to be exhaustive.

### Other possible sources of error

Other potential sources of experimental variation contributing to over- or underlabeling appeared to be minimal. However, since some of these strictly technical factors have been mentioned by others as possible sources of errors in experiments of this kind (e.g., Kim and Strick, '76; Adams, '77, '80; Rosene and Mesulam, '78; Mesulam, '82), they were not disregarded before their specific variation was carefully measured in several of the preliminary test cases. Perhaps it is worth repeating that in each test case, the procedure which was eventually adopted as the standardized one yielded as many or more labeled neurons than any other procedure employed. Therefore, there is reason to believe that among the array of retrograde tract-tracing techniques available at the time, the number of labeled neurons may be maximized by the cut-and-fill with HRP technique, although no claim for complete labeling can be made.

Finally, because the large number of sections examined necessitated a section-sampling procedure, it is possible that some cells were labeled but were overlooked. For example, a small number of amygdalospinal neurons have

previously been reported in macaque and cat (Mizuno et al., '85; Sandrew et al., '86). However, in the present materials, amygdalospinal neurons were seen only in some insectivores, in the short-tailed opossum, and armadillo, but not in macaque, cat, or any of the other 15 species. Since such discrepancies between the present results and previously published reports were very rare and in the few cases where they did occur, very small numbers of neurons or very sparsely populated nuclei were involved, they would seem to be more of a minor distraction than a serious contradiction. When compared with the large number of reliably and vividly labeled tracts seen in all the animals of the present study and in most if not all of the same species reported by others, we certainly do not wish to claim that a given tract does not exist if it has been reported by others but not seen here. We will return to this point below.

### Basis for interspecies comparisons of descending spinal pathways

It is a tenet of comparative neurology that no single criterion is sufficient to establish homologies among neuronal populations (Campbell and Hodoss, '70). To the contrary, almost every neuronal characteristic now known is at least a defensible candidate for inclusion among the criteria for homology. Of course, it is beyond the scope of any single experimental study to compare each cell group across every species on each of the myriad defensible criteria: the present study is no exception. However, in the present study, an unusually large number of characteristics of the labeled cell groups were available for use in their identification by virtue of the vivid labeling obtained. Each of these characteristics was rigorously applied in order to establish proper nuclear identifications (Fig. 9). As it turned out, every characteristic reinforced the identification based on the other characteristics with very few exceptions. That is, there were very few arguable identifications in any species and none in the 27 tracts listed as common to all of the species. In the rare cases where equivocation did occur, we have cited it in the description of the cell group.

Therefore, though perfect assurance of true homology is never possible, the present materials allowed a relatively high degree of confidence in the identification of the labeled cell groups. Nevertheless, there remain several reasons why the conclusions of the present study certainly cannot be considered to end the question of commonalities among the descending pathways.

### Adequacy of sample

After having satisfied cytoarchitectonic criteria for similarities in morphological detail, it remains to be illustrated that each of the pathways probably arose from a common ancestral precursor (Ghiselin, '76). The best way to attain this goal (short of studying every species) is with a random sample of species whose ancestral histories are certain (Fisher, '25).

Although we examined 22 diverse species, they do not constitute a random sample. Some major groups—for example, Cetacea, Chiroptera, and Artiodactyla—are not represented. Representatives of each of these large orders would be expected in a truly random sample of 22 species of mammals. Therefore, generalizations to the entire class of Mammalia even of the characteristics found common to all of the 22 species embodies some extra statistical risk. However, because we chose the animals in the present sample based not upon prior reports of their descending systems but upon paleontological, taxonomic, and compar-

ative anatomical grounds for their sequential kinship with mankind (as well as their availability, body size, cost, and other practical considerations), statistical calculation shows that the sample is a good one for the evolutionary goal for which it was intended and at least an adequate one for more wider comparative purposes.

The question of sample "adequacy" turns on the possibility of mistakenly concluding homology (sameness due to ancestry) when what has been seen is merely homoplasy (sameness due to anything except ancestry). The probability of this mistake cannot be calculated exactly, but its upper limit, or worst case, can be readily estimated.

For example, if the probability of a given tract arising spontaneously in any one species (i.e., being "homoplastic" or "derived" without regard to immediately prior ancestry) were arbitrarily presumed to be 50% (an intentionally high and absurd value), then the probability that the tract would be found in all 22 species (and thus, homoplasy mistaken for commonality and homology) is  $(0.5)^{22}$ , which is less than one in a million. Therefore, with even an absurdly high probability of homoplasy at the species level, there is virtually no chance that any one of the 27 pathways would arise in every species in the sample.

Pursuing this line of reasoning still further, this time skipping to the ordinal level and presuming that the probability of a given tract arising spontaneously (or homoplastically) in any one order is, again, a very high 50%, then the probability of seeing the same tract in each of the eight orders represented here is still a very small  $(0.5)^8$ , which is less than one in 256. Therefore, the large and varied sample used here, despite its nonrandomness, serves the purpose of decreasing the probability of a serious error in inference: there is little chance that homoplasy has been mistaken for homology at any taxonomic level below subclasses (i.e., marsupial vs. placental).

Finally, if instead of presuming a 50% chance of homoplasy, one presumes a more reasonable and empirically based value of  $4/31$  (the number of tracts not common to all species/the total number of identified tracts), then the probability of mistaking homoplasy for homology among the two subclasses is  $(4/31)^2$ , among the eight orders is  $(4/31)^8$ , and among the 22 species is  $(4/31)^{22}$ , which are less than two in 100, one in 100 million, and one in septillion, respectively. Therefore, short of a long and unrelenting string of bad luck, it is reasonably safe to conclude that each of the 27 pathways found to be common to all 22 species, 19 genera, etc., in the present study was certainly common to the extinct animals in mankind's mammalian lineage, is almost certainly common to all placental mammals, and is probably common to all viviparous mammals as well.

However, despite the statistical assurance that the 27 common pathways are probably common to therian mammals, the inverse of this conclusion has little or no assurance, statistical or otherwise. That is, it is false to conclude that because a particular tract was not labeled in every species, it is not present in most or even in all mammals. Both because of the possible existence of retrograde-labeling phobias of particular tracts in particular species discussed above and because of a lack of a reasonable estimate of the probability of a given tract becoming vestigial at some point in a species' lineage, the nonlabeling of a tract in a species, as opposed to the nonexistence of the tract in that species, defies disentanglement. Therefore, the most conservative conclusion at present is that there exist 27 pathways common to all viviparous mammals and possibly several more.

### Implications for the phylogenetic history of descending spinal pathways in mammals

The results show that aside from a few and apparently minor exceptions, a list of supraspinal cell groups originating descending pathways to the spinal cord is strikingly similar in each of the 22 species represented here. For the reasons outlined already, this uniformity among diverse species suggests that the 27 common pathways are probably homologs among all viviparous mammals and were also present among the extinct mammals in mankind's ancestral lineage.

However, this is not to say that these common tracts are identical in every detail. For example, in the four cell groups (tectum, cerebral cortex, red nucleus, and hypothalamus) where the numbers of labeled cells were counted in each of the species, quantitative variation was found even among species within the same genera, larger variation was found among genera within the same family, still larger variation among families within the same order, etc. This specific, generic, familial, and ordinal variation and its relationship both to ancestry and to current ecology are the subject of future reports. For the present purpose, however, the striking qualitative similarities among the species suggest that a "mammalian common plan" consisting of at least these 27 descending spinal pathways was probably present before the common ancestry of the 22 species, and therefore very early in the evolutionary history of mammals. Furthermore, at least these 27 pathways must have been preserved throughout the later periods of speciation at least along the 22 lineages represented. On the other hand, the great quantitative differences in the tracts among the species imply that the absolute and relative number of fibers within each of the tracts must have been responsive to selective pressures sufficiently strong or sufficiently persistent to wreak measurable if not substantive changes even among the most closely related species.

### Implications for the phylogenetic history of descending spinal pathways in vertebrates

Because the 27 pathways are probably common to marsupials and placentals alike, it follows that these pathways probably originated before the last common ancestor of marsupial and placental mammals—that is, in or before the late Jurassic, perhaps 145 million years ago (Lillegraven et al., '79).

This conclusion, holding equally for each of the 27 tracts, leads to the notion that some of these tracts might have originated still earlier—that is, they might have arisen among the nonmammalian vertebrates in the ancestral lineage eventually giving rise to the mammals. Indeed, if evidence for each of the pathways is sought in studies of nonmammalian vertebrates, it becomes apparent that many of the pathways may have originated much earlier than the relatively recent appearance of true mammals—some possibly dating to the origin of vertebrates themselves.

### The vertebrate plan

Even in the most primitive vertebrate forms still extant, neurons in the spinal cord are known to be influenced by axons originating at supraspinal levels. The application of retrograde transport techniques such as those used here, have already demonstrated supraspinal origins of spinal fibers in a wide variety of vertebrates: e.g., see Ronan ('81), Ronan and Northcutt ('81) in Agnatha; Cruce and Northcutt ('85); Smeets and Timerick ('81) in Chondrichthyes;

TABLE 2. Major Sources of Descending Spinal Pathways in Vertebrates<sup>1</sup>

| Class          | Reticular form. | Vestibular nuclei | Interstitial n. | Spinal n. V | Raphe complex | n. solitary tract | Red n. | Hypothalamus | Cerebellar nuc. | Locus coeruleus | Oculomotor complex | Dorsal column n. | Pallium |
|----------------|-----------------|-------------------|-----------------|-------------|---------------|-------------------|--------|--------------|-----------------|-----------------|--------------------|------------------|---------|
| Agatha         | +               | +                 | +               |             |               |                   |        |              |                 |                 |                    |                  |         |
| Chondrichthyes | +               | +                 | +               | +           | +             | +                 | ±      | +            | —               | +               | +                  |                  | —       |
| Osteichthyes   | +               | +                 | +               | +           | +             | +                 | +      | +            | —               | ?               |                    |                  | —       |
| Amphibia       | +               | +                 | +               | +           | +             | +                 | ±      | +            | +               | ?               |                    |                  | —       |
| Reptilia       | +               | +                 | +               | +           | +             | +                 | ±      | +            | +               |                 | +                  | +                | —       |
| Mammalia       | +               | +                 | +               | +           | +             | +                 | +      | +            | +               | +               | +                  | +                | +       |
| Aves           | +               | +                 | +               | +           | +             | +                 | +      | +            | +               | +               | ?                  | +                |         |

<sup>1</sup>+, labeled neurons identified which project to the spinal cord. —, retrogradely labeled neurons sought, but not identified. ±, retrogradely labeled neurons identified in some studies, but not others; differences may or may not be systematic. ?, retrogradely labeled cells identified in general vicinity of cell group, but homology with other classes still in question. Note: Several additional minor cell groups (nuc. ambiguus, parabrachial, perihypoglossal, motor n. of X, periaqueductal grey, cuneiform nuc., nuc. of posterior commissure, mesencephalic nuc. of V, optic tectum, torus semicircularis, etc.) send axons to the spinal cord in species representing one or more vertebrate classes but are not included here.

Demski and Sloan ('85); Gregory and Tweedle ('85); Kimmel et al. ('82); Northcutt ('83); Oka et al. ('86); Prasada Rao et al. ('87); Ronan and Northcutt ('85) in Osteichthyes; Forehand and Farel ('82); Grover and Grusser-Cornehls ('80); Kokoros and Northcutt ('77); ten Donkelaar et al. ('81), Toth et al. ('85) in Amphibia; Bruce et al. ('80), Newman et al. ('83), ten Donkelaar et al. ('80), Wolters et al. ('82), Woodson and Kunzle ('82) in Reptilia; Basbaum and Fields ('79), Burton and Loewy ('77), Carlton et al. ('85), Coulter et al. ('79), Kuypers ('82), Martin et al. ('81) and many others in Mammalia; and Berk and Finkelstein ('83), Cabot et al. ('82), Gross and Oppenheim ('85), Wild et al. ('79), Wold ('78) in Aves.

Table 2 summarizes these results in other major vertebrate classes for only those descending pathways found to be present in mammals. Although these data do not provide a perfectly complete picture of all descending spinal afferents, and the species which have been studied cannot be considered to be either perfect representatives or random samples of their entire class, the many cases already available display several clear similarities which, once again, short of an improbable run of bad luck, shed light on the very early, premammalian, history of those descending pathways now found in mammals.

As shown in Table 2, at least three major supraspinal cell groups have projections to the spinal cord in every species of vertebrate studied to date: the reticular formation (reticulospinal tract), the vestibular nuclei (vestibulospinal tract), and the interstitial nucleus of the medial longitudinal fasciculus. Further, in all vertebrates yet studied, the spinal projections of each of these three cell groups travel down the cord in the ventral funiculus and each terminates in the ventromedial spinal gray (e.g., see ten Donkelaar, '76).

Based upon their similar origins, similar trajectories and similar terminations, it is probably safe to conclude that each of these three pathways is probably a true homolog of its respective counterpart throughout the vertebrate subphylum. If so, they can be considered to constitute a "vertebrate common plan" of descending spinal pathways (also see ten Donkelaar, '82).

### The jawed-vertebrate (or gnathostome) augmentation of the vertebrate common plan

Table 2 also shows that in all vertebrate classes except Agnatha (i.e., excepting only the jawless fishes such as lamprey and hagfish), five other cell groups originating descending spinal fibers (nucleus of V or trigeminospinal

tract, raphe nucleus or raphe-spinal tract, nucleus solitarius or solitariospinal tract, red nucleus or rubrospinal tract, and hypothalamus or hypothalamospinal tract) have been described.

The fibers originating from each of these cell groups also follow a similar spinal course and each has a similar termination pattern across the jawed-vertebrate or gnathid classes, supporting the notion that they too are authentic homologs with their respective counterparts. Inasmuch as these pathways can, in principle, be traced back to a common, primitive, jawed, perhaps chondrichthyan ancestor but not further, it is reasonable to conclude that these pathways represent an early "gnathid augmentation" of the vertebrate common plan. (Although there is still some question concerning their status as homologs in anamniotic vertebrates, two additional cell groups, locus coeruleus and the nucleus of Edinger-Westphal, may also project to the spinal cord in all gnathid vertebrates—see queries in Table 2 and ten Donkelaar et al., '81.)

In addition to the five major cell groups shown in Table 2, a sixth cell group located in the ventral thalamus (which includes zona incerta) may also originate descending spinal fibers in all jawed-vertebrate classes.

### The tetrapod augmentation

Table 2 shows that the cerebellum (i.e., the deep cerebellar nuclei) projects to the spinal cord in amphibians, reptiles, mammals, and birds. Again, since this cerebellospinal tract has a similar origin, trajectory, and termination pattern, this tract is probably homologous across the tetrapod classes. If so, the basic vertebrate plan appears to have undergone a second augmentation with the appearance of tetrapodal vertebrates, true amphibians.

In addition, cell groups in the subpallium (or "ventral striatum" or "basal telencephalon") have been described which originate descending spinal fibers in amphibians and reptiles (Bruce et al., '80; ten Donkelaar et al., '81; Toth et al., '85). It is possible that these subpallial-spinal fibers of amphibians and reptiles correspond to spinal-projecting neurons located in the amygdala or located ventral to the anterior commissure of one or more species of mammals (Mizuno et al., '85; Sandrew et al., '86; and present results). If so, these subpallial-spinal projections might also be part of the tetrapod augmentation.

### The amniote augmentation

Table 2 shows that at least one group of cells found in the dorsal column nuclei (especially the cuneate nucleus, the origin of the cuneospinal tract) has been found to project to

the cord in mammals, reptiles, and birds but so far not in amphibians, osteichthyeans, chondrichtheans, or agnathans. Once more, since the cuneospinal tract can be traced to a common reptilian ancestor but not further, these pathways might represent a third, "amniotic" augmentation of the vertebrate plan.

### The reptile-mammal divergence and the mammalian augmentation

Table 2 shows that axons originating in *pallial* cell groups seem to project to the spinal cord in only one vertebrate class, Mammalia. Therefore, it appears that the vertebrate common plan was augmented still a fourth time in the lineage leading to mammals by fibers originating in neocortex. Of course, this unique and possibly sole major addition in the mammalian lineage, a true corticospinal tract, eventually became more massive than any other descending pathway.

### SUMMARY

The results show that at least 27 pathways of supraspinal origin descend to the spinal cord in each of the 22 species represented in the study. On the basis of this qualitative commonality together with the size and diversity of the sample of species, it can be concluded that these 27 pathways are probably common to all viviparous (therian) mammals and were also common to mankind's prehuman mammalian ancestors.

Comparison of the present results with similar experimental findings in other vertebrate classes further suggests that several of these pathways can be traced to even more distant, premammalian, ancestry, with at least three possibly as old as the entire vertebrate subphylum.

The phylogenetic history of direct descending projections from brain to spinal cord along the ancestral lineage leading to mammals seems to have consisted of a series of widely spaced, steplike augmentations of the previous complement of descending tracts with one and only one major descending pathway, the corticospinal tract, being uniquely mammalian.

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